MATH 74: ALGEBRAIC TOPOLOGY

PRISHITA DHARAMPAL

1. Topological Spaces & Homeomorphisms

Definition 1.1. *A **topology** on a set X is a collection J of subsets of X having the following properties.*

(1) $\emptyset, X \in J$
(2) The union of the elements of any sub-collection of J is in J.
(3) The intersection of finitely many elements in J is in J.

A set for which a topology has been specified is said to be a topological space. We think of it as a pair (X, J).

Definition 1.2. *Elements of J are called **open sets** of the topological space (X, J).*

Definition 1.3. *A subspace Y of a topological space (X, J) is a subset $Y \subset X$ which is given the **subspace topology** where the open sets of $Y = \{U \cap Y : U \in J\}$*

Definition 1.4. *A space X is **path connected** if every pair of points of X can be joined by a path in X.*

Definition 1.5. *Let (X_1, J_1) and (X_2, J_2) be topological spaces. A map $f : (X_1, J_1) \to (X_2, J_2)$ is said to **continuous** if $f^{-1}(U) \in J_1$ for all $U \in J_2$.*

Definition 1.6. *Let (X, J) and (X', J') be topological spaces. Then they are **homeomorphic** if $\exists$ a continuous bijection $f : (X, J) \to (X', J')$. So that f^{-1} is also continuous.*

Example 1.1. *Show that any two open intervals of $\mathbb{R}$ are homeomorphic (where we put subspace topology on the open intervals).*

Solution. Let (a, b) and (c, d) be open intervals and define $f : (a, b) \to (c, d)$ as follows:

$$f(x) = c + (x - a)\frac{d - c}{b - a}$$

Then, f is continuous since it is a linear function in x, bijective because it is strictly increasing (injective) and has a continuous inverse.

Remark 1.7. *Number of path-connected topologies is preserved by continuous maps. For instance, $[0, 1]$ is not homeomorphic to $(0, 1)$, since $[0, 1] \setminus \{0\} = (0, 1]$ is path-connected but $(0, 1) \setminus \{pt\} = (0, pt) \cup (pt, 1)$ which has two path-connected components. Similarly, $\mathbb{R}$ is not homeomorphic to $\mathbb{R}^2$ since $\mathbb{R} \setminus \{pt\}$ gives us two disconnected components, but $\mathbb{R}^2 \setminus \{pt\}$ is path-connected.*

Example 1.2.

(a) Is $\mathbb{R}^2$ homeomorphic to $\mathbb{R}^3$?
(b) Is $S^2 (= x^2 + y^2 + z^2 = 1)$ homeomorphic to $T^2 = S^1 \times S^1$?
(c) Is S^2 homeomorphic to $S^1 \times [a, b)$?

Solution.

(a) No, since $\pi_1(\mathbb{R}^2 \setminus \{\text{pt}\}) = \pi_1(S^1) = \mathbb{Z}$ and $\pi_1(\mathbb{R}^3 \setminus \{\text{pt}\}) = \pi_1(S^2) = 0$
(b) No, since $\pi_1(S^2) = 0$ and $\pi_1(T^2) = \mathbb{Z} \oplus \mathbb{Z}$.
(c) No, since $\pi_1(S^2) = 0$ and $\pi_1(S^1 \times [a,b)) = \pi_1(S^1) = \mathbb{Z}$.

Definition 1.8. *If f and f' are continuous maps from spaces X to Y we say that f is homotopic to f' if $\exists$ a continuous map $F : X \times [0,1] \to Y$ such that $F(x,0) = f(x)$ and $F(x,1) = f'(x)$. F is called the* ***homotopy*** *between f and f', and we write $f \simeq f'$.*

Definition 1.9. *If X and Y are topological spaces, we can define a* ***product topology*** *on $X \times Y$ by saying that a basis consists of the subsets $U \times V$ as U ranges over open sets in X and V ranges over open sets in Y.*

Definition 1.10. *Define a* ***path*** *f as $f : [0,1] \to X$ with $f(0) = x_0$ and $f(1) = x$.*

Definition 1.11. *Two paths $f, f' : [0,1] \to X$ are said to be path homotopic if there exist a continuous map $F : I \times I \to X$ such that $\forall s \in I$,*

$$F(s,0) = f(s), \qquad F(s,1) = f'(s)$$

and $\forall t \in I$,

$$F(0,t) = x_0, \qquad F(0,t) = x$$

Then we write $f \underset{p}{\simeq} f'$.

Example 1.3. *Let $f : X \to \mathbb{R}^2$ and $g : X \to \mathbb{R}^2$ then $f \simeq g$.*

Proof. Want to find F such that $F : X \times I \to \mathbb{R}^2$, $F(x,0) = f(x)$, and $F(x,1) = g(x)$ for all $x \in X$. The following map (straight-line homotopy) works.

$$F(x,t) = (1-t)f(x) + tg(x)$$

□

Lemma 1.12. *The relations $\simeq$ and $\simeq_p$ are equivalence relations.*

Proof.

Homotopy is an equivalence relation:

(1) **Reflexive:** $f \simeq f$ by $F(s,t) = f(s)$.
(2) **Symmetric:** Let $f \simeq g$ by the homotopy $F(s,t)$. Then $g \simeq f$ is homotopic by the homotopy $G(s,t) = F(s, 1-t)$.
(3) **Transitive:** Let $f \simeq g$ by $F(s,t)$ and $g \simeq h$ by $G(s,t)$. We can show that $f \simeq h$ by

$$H(s,t) = \begin{cases} F(s,2t), & 0 \le t \le 1/2 \\ G(s,2t-1), & 1/2 \le t \le 1 \end{cases}$$

Then $H(s,0) = F(s,0) = f(s)$, and $H(s,1) = G(s,1) = h(s)$. Hence, $f \simeq h$.

Path-Homotopy is an equivalence relation:

(1) **Reflexive:** $f \simeq_p f$ by $F(s,t) = f(s)$.

(2) **Symmetric:** Let $f \simeq_p g$ by the homotopy $F(s,t)$. Then $g \simeq_p f$ is homotopic by the homotopy $G(s,t) = F(s, 1-t)$.
(3) **Transitive:** Let $f \simeq_p g$ by $F(s,t)$ and $g \simeq_p h$ by $G(s,t)$. We can show that $f \simeq_p h$ by,

$$H(s,t) = \begin{cases} F(s,2t), & 0 \leq t \leq 1/2 \\ G(s,2t-1), & 1/2 \leq t \leq 1 \end{cases}$$

Where
- $H(s,0) = F(s,0) = f(s)$
- $H(s,1) = G(s,1) = h(s)$
- $H(0,t) = x_0$ and $H(1,t) = x_1$.

Hence, $f \simeq_p h$.

Thus path-homotopies are equivalence relations and we write $[f]$ to denote the path-homotopy equivalence class of f. □

Lemma 1.13 (Pasting Lemma). *Let $X = A \cup B$ be a topological space with $A, B \subset X$ are closed. Let $f : A \to Y, g : B \to Y$ be continuous and $f(x) = g(x)$, $\forall x \in A \cap B$, then*

$$h(x) = \begin{cases} f(x), \, x \in A \\ g(x), \, x \in B \end{cases}$$

is continuous.

Definition 1.14. *Let $f : I \to X$ be a path from x_0 to x_1 and $g : I \to X$ be a path from x_1 to x_2, then we define a new path*

$$f \star g : I \to X$$

from x_0 to x_2 by

$$f \star g(t) = \begin{cases} f(2t), \, t \in [0, 1/2] \\ g(2t-1), \, t \in [1/2, 1] \end{cases}$$

*where $\star$ is **composition of paths**.*

Some Properties of '$\star$'.

(1) '$\star$' is well-defined operation in the path homotopy equivalence class,

$$[f] \star [g] = [f \star g]$$

(2) Identity:

$$[f] \star [e_{x_1}] = [f], \quad [e_{x_0}] \star [f] = [f]$$

where e_{x_t} is the constant path based at $x \in X$ and $t \in [0,1]$

(3) Inverse:

$$[f] \star [\overline{f}] = [e_{x_0}], \quad [\overline{f}] \star [f] = [e_{x_1}]$$

where $\overline{f}(t) = f(1-t)$.

(4) Associative:

$$[f] \star ([g] \star [h]) = ([f] \star [g]) \star [h]$$

Proof.

(1) Let $f \simeq_p f'$ by F and $g \simeq_p g'$ by G, then we can show that $f \star g \simeq_p f' \star g'$ by defining $H : I \times I \to X$ such that

$$H(s,t) = \begin{cases} F(2s,t),\ s \in [0,1/2] \\ G(2s-1,t),\ s \in [1/2,1] \end{cases}$$

(2) $f \star e_{x_1} \simeq_p f$ by $H : I \times I \to X$ such that

$$H(s,t) = \begin{cases} f\left(\frac{2 \cdot s}{2-t}\right),\ s \in \left[0, \frac{2-t}{2}\right] \\ x_1,\ s \in \left[\frac{2-t}{2}, 1\right] \end{cases}$$

(3) To see $e_{x_0} \simeq_p f \star \overline{f}$ define a path $\alpha(t)$ that starts at x_0 and ends at $f(t)$. Then the path-homotopy is given by $\alpha(t) \star \overline{\alpha}(t)$.

$$H(s,t) = \begin{cases} f(2st),\ s \in [0,1/2] \\ f(2t(1-s)),\ s \in [1/2,1] \end{cases}$$

(4) By well-definedness,

$$[f] \star ([g] \star [h]) = [f] \star [g \star h] = [f \star g \star h] = [f \star g] \star h = ([f] \star [g]) \star [h]$$

$\square$

Remark 1.15. *Note that $\star$ operation on the set of path-homotopy equivalence classes of paths in the space X forms a groupoid structure.*

2. The Fundamental Group

Definition 2.1. *Let X be a space and $x_0 \in X$ be a point in X we say that a path $f : I \to X$ is a **loop** based at x_0 if $f(0) = f(1) = x_0$*

Definition 2.2. *The set of all path-homotopy equivalence classes of loops based at x_0 forms a group under $\star$ called the **fundamental group** of x based at x_0, denoted as $\pi_1(X, x_0)$*

- *Identity:* $[e_{x_0}]$
- *Inverses:* $[f] \to [\overline{f}]$

Example 2.1. $\pi_1(\mathbb{R}^n, (0,0)) \equiv 0 \equiv \langle e_{origin} \rangle$

Proof. Take any two loops $f(t), g(t)$ based at the origin. Then consider the straight-line homotopy,

$$H(s,t) = f(s)(1-t) + tg(s)$$

where,

- $H(s,0) = f(s)$
- $H(s,1) = g(s)$
- $H(0,t) = f(0)(1-t) + tg(0) = (0,0) = H(1,t)$

Hence, it is a path-homotopy. □

Definition 2.3. *Let α be a path from x_0 to x_1, where $x_0, x_1 \in X$. We define a map $\widehat{\alpha} : \pi_1(X, x_0) \to \pi_1(X, x_1)$ such that*

$$\widehat{\alpha}([f]) := [\overline{\alpha}] \star [f] \star [\alpha] = [\overline{\alpha} \star f \star \alpha]$$

Proposition 2.4. *$\widehat{\alpha}$ is well-defined and an isomorphism.*

Proof. Let $f \simeq_p f'$. We need to check that

$$\begin{aligned} [\overline{\alpha}] \star [f] \star [\alpha] &= [\overline{\alpha}] \star [f'] \star [\alpha] \\ \iff [\overline{\alpha} \star f \star \alpha] &= [\overline{\alpha} \star f \star \alpha] \\ \iff \overline{\alpha} \star f \star \alpha &\simeq_p \overline{\alpha} \star f' \star \alpha \end{aligned}$$

Since $\star$ is well-defined, $[f] \star [\alpha] = [f'] \star [\alpha]$, $\widehat{\alpha}$ is well-defined

Let $[f], [g] \in \pi_1(X, x_0)$.

$$\begin{aligned} \widehat{\alpha}([f]) \star \widehat{\alpha}([g]) &= [\overline{\alpha}] \star [f] \star [\alpha] \star [\overline{\alpha}] \star [g] \star [\alpha] \\ &= [\overline{\alpha}] \star [f] \star [e_{x_0}] \star [g] \star [\alpha] \\ &= [\overline{\alpha}] \star [f] \star [g] \star [\alpha] \\ &= \widehat{\alpha}([f] \star [g]) \end{aligned}$$

Hence $\widehat{\alpha}$ is a homomorphism. To check that $\widehat{\alpha}$ is an isomorphism, we see that $\widehat{\overline{\alpha}}(\widehat{\alpha})([f]) = [f]$ for all $[f] \in \pi_1(X, x_0)$. □

Corollary 2.4.1. *The fundamental group of X based at x_0 only depends on the path-connected component of $x_0 \in X$ upto isomorphism.*

Definition 2.5. *Let $h : X \to Y$ be a continuous map such that $h(x_0) = y_0$, $x_0 \in X, y_0 \in Y$. Then define $h_\star : \pi_1(X, x_0) \to \pi_1(Y, y_0)$ by $h_\star([f]) = [h \circ f]$. This is a **push-forward** map.*

Proposition 2.6. *$h_\star$ is a well-defined homomorphism.*

Proof. Let $f \simeq_p f'$. We need to check that $h \circ f \simeq_p h \circ f'$. Let $H : I \times I \to X$ be the path homotopy between f and f'. Consider $h \circ H : I \times I \to Y$.

$$\begin{aligned} h \circ H(s, 0) &= h \circ f \\ h \circ H(s, 1) &= h \circ f' \\ h \circ H(0, t) &= h \circ f(0) \\ h \circ H(1, t) &= h \circ f(1) = h \circ f(0) \end{aligned}$$

Hence $h \circ H$ is the required path-homotopy.

Consider $[f], [g] \in \pi_1(X, x_0)$. We want to show that $h_\star([f]) \star h_\star([g]) = h_\star([f] \star [g])$.

$$h_\star([f]) \star h_\star([g]) = [h \circ f] \star [h \circ g]$$

Note that it suffices to show that $h(f \star g) = (h \circ f) \star (h \circ g)$. Now,

$$\begin{aligned} h(f \star g)(s) &= \begin{cases} (h \circ f)(2s), & s \in [0, \frac{1}{2}] \\ (h \circ g)(2s - 1), & s \in [\frac{1}{2}, 1] \end{cases} \\ &= (h \circ f) \star (h \star g) \end{aligned}$$

Hence proved. $\square$

Theorem 2.7. *If $h : (X, x_0) \to (Y, y_0)$ and $k : (Y, y_0) \to (Z, z_0)$ then*

(a) $(k \circ h)_\star = k_\star \circ h_\star$

(b) *If* $\mathrm{id} : (X, x_0) \to (X, x_0)$ *is the identity map then* $(\mathrm{id})_\star$ *is the identity* $\pi_1(X, x_0) \to \pi_(X, x_0)$.

Proof.

(a) We have,

$$\begin{aligned} (k \circ h)_\star([f]) &= [(k \circ h) \circ (f)] \\ &= [k(h(f))] \\ &= [k_\star(h_\star([f]))] \\ &= (k_\star \circ h_\star)([f]) \end{aligned}$$

(b) $(\mathrm{id})_\star[f] = [\mathrm{id} \circ f] = [f]$

$\square$

Corollary 2.7.1. *If $h : X \to Y$ is a homeomorphism then $\pi(X, x_0) \cong \pi(Y, h(x_0)$ via $h_\star$.*

Definition 2.8. *Let $P : E \to B$ be a surjective map. A open set $U \subseteq B$ is said to be **evenly covered** by P if $P^{-1}(U) = \bigsqcup_\alpha V_\alpha$, where V_α are open sets of E for all α and $P|_{V_\alpha}$ is a homeomorphism of V_α onto U.*

Definition 2.9. *Let $P : E \to B$ be surjective if every point $b \in B$ has a neighbourhood $U \subset B$ such that U is evenly covered by P. Then we say that P is a **covering map** and E is a **covering space** of B.*

Example 2.2.

(1) $\mathrm{id} : X \to X$.
(2) $X \times \{1, 2, \cdots, n\} \xrightarrow{P} X$ *where* $P(x, i) = x$.

Theorem 2.10. *The map $P : \mathbb{R} \to S^1$ such that $x \mapsto (\cos(2\pi x), \sin(2\pi x))$ is a covereing map.*

Proof. Let $U_1 = S^1 \setminus \{(1, 0)\}$ and $U_2 = S^1 \setminus \{(-1, 0)\}$. Then it suffices to show that both of these sets is evenly covered. We have,

$$P^{-1}(U_1) = \sqcup_{n \in \mathbb{Z}} (n, n+1) =: V_n$$

And,

$$P^{-1}(U_2) = \sqcup_{n \in \mathbb{Z}} \left(n - \frac{1}{2}, n + \frac{1}{2} \right) =: W_n$$

Note that P is injective on any interval of length less than 1, because if $P(x) = P(y)$ then $(\cos(2\pi x), \sin(2\pi x)) = (\cos(2\pi y), \sin(2\pi y)) \iff x - y \in \mathbb{Z}$ but that is not possible if $x - y < 1$. $P|_{V_n}$ is surjective onto U_1 by IVT and is continuous because P is continuous. It has a continuous inverse since it is a continuous bijection from an open interval onto an open subset of S^1.

Also, the invervals $(n, n+1)$ are pairwise disjoint for each n. A similar argument applies to W_n. Hence, both U_1 and U_2 are evenly covered and P is a covering map. $\square$

Definition 2.11. *If a path-connected space X has $\pi_1(X, x_0) \cong 0$ then we say that X is **simply-connected**.*

Lemma 2.12. *If X is simply-connected, then any two paths in X with the same initial point and end point are path-homotopic.*

Proof. $f \star \overline{g} \simeq_p e_{x_0}$ and $f \simeq_p f \star e_{x_1} \simeq_p f \star (\overline{g} \star g) \simeq_p (f \star \overline{g}) \star g \simeq_p e_{x_0} \star g \simeq_p g$. $\square$

Definition 2.13. *Let $p : E \to B$ be a surjective map, and $f : X \to B$ be a map. Then a map $\widetilde{f} : X \to E$ is called **lifting** of f to E if $p \circ \widetilde{f} = f$.*

$$\begin{tikzcd} & E \arrow[d, "p"] \\ X \arrow[ur, dashed, "\widetilde{f}"] \arrow[r, "f"'] & B \end{tikzcd}$$

Lemma 2.14 (Path Lifting Lemma)**.** *Let $p : E \to B$ be a covering map with $p(e_0) = b_0$. Any path $f : [0, 1] \to B$ beginning at b_0 can be lifted uniquely to a path $\widetilde{f} : [0, 1] \to E$ beginning at e_0.*

Proof. Cover B by open sets U, each of which is evenly covered by p. We can find a subdivision (partition) defined by $s_0, \ldots, s_n \in [0, 1]$ such that $f([s_i, s_{i+1}])$ is contained entirely inside a evenly covered U for all i. Now we will define $\widetilde{f}$ inductively. Suppose that $\widetilde{f}(0) = e_0$ and it has been defined for $0 \le s \le s_i$ for some i. We will now define $\widetilde{f}$ at $[s_i, s_{i+1}]$. Suppose that $f([s_i, s_{i+1}]) \subset U$ where U is evenly covered open set of B. This implies that $p^{-1}(U) = \sqcup_\alpha V_\alpha$. Now, $\widetilde{f}(s_i)$ lies in one such V_α say V_0.

Define $\widetilde{f}$ on $[s_i, s_{i+1}]$ as $\widetilde{f}(s) := (p|_{V_0})^{-1}(f(s))$ for all $s \in [s_i, s_{i+1}]$. By pasting lemma $\widetilde{f}$ is continuous on $[0, s_{i+1}]$ and we argue inductively to define $\widetilde{f}$ for $[0,1]$.

By induction, we can again suppose that for another lift $\widetilde{f}' : [0,1] \to E$ of f with $\widetilde{f}'(0) = e_0$, it agrees with $\widetilde{f}$ of $0 \le s \le s_i$ for some i

$$\widetilde{f}'(s) = \widetilde{f}(s), \qquad \forall s \in [0, s_i]$$

Now we would like to show that

$$\widetilde{f}'(s) = \widetilde{f}(s), \qquad \forall s \in [s_i, s_{i+1}]$$

$$\widetilde{f}'(s) = (p|_{V_0})^{-1} \circ f(s), \qquad \forall s \in [s_i, s_{i+1}]$$

$\widetilde{f}'(s_i)$ lies in V_0 since $\widetilde{f}'(s_i) = \widetilde{f}(s_i)$. $\implies$ $\widetilde{f}'([s_i, s_{i+1}])$ lies in V_0.

$p \circ \widetilde{f}' = f$ on $[s_1, s_{i+1}]$. But $\widetilde{f}'([s_i, s_{i+1}]) \subset V_0$ and $P|_{V_0}$ is a homeomorphism.

Then for all $s \in [s_i, s_{i+1}]$

$$\implies \widetilde{f}'(s) = (p|_{V_0})^{-1} \circ f(s) = \widetilde{f}(s)$$

□

Lemma 2.15 (Path Homotopy Lifting Lemma)**.** *Let $p : E \to B$ be a covering map. Let $p(e_0) = b_0$. Let $f : I \times I \to B$ be a map, with $f(0,0) = b_0$. Then there exists a lifting of f to E such that $\widetilde{f}(0,0) = e_0$. If f is a path homotopy, so is $\widetilde{f}$.*

Proof. **TODO:** $I \times I$ is a rectangle. Pick a parition of both sides, then the image of these small rectangles will be contained in U

1. Define $\widetilde{f}$ on $0 \times I$ and $I \times 0$ by lifting both of those paths to paths that begin at e_0.

2. Take a subdivision of $I \times I$ say $I_i \times J_j$ such that each of those small rectangles are contained in an evenly covered open set of B.

3. By induction assume that $\widetilde{f}$ has been defined on $I_i \times J_j$ such that $i < i_0, j = j_0$.

4. Now, to define $\widetilde{f}$ on $I_{i_0} \times I_{j_0}$ we consider the image of the union of left and bottom edge of $I_{i_0} \times I_{j_0}$ under $\widetilde{f}$. This lies inside $V_0 \subset p^{-1}(U)$. □

Remark 2.16. *The lifted path homomoty is also unique.*

Lemma 2.17. *Let $P : E \to B$ be a covering map with $P(e_0) = b_0$. Let f, g be two paths in B from b_0 to b_1. Let $\widetilde{f}, \widetilde{g}$ be their lifts beginning at e_0. Then if $f \simeq_p g \implies \widetilde{f}$ and $\widetilde{g}$ end at the same point and they are path homotopic.*

Proof. Assume $f \simeq_p g$ with the path homotopy $F : I \times I \to B$. Then Lemma 2.15 implies that F lifts to $\widetilde{F} : I \times I \to E$ with $\widetilde{F}(0,t) = e_0$.

Then we can say that $\widetilde{F}(1,t) = e_1$ for some $e_1 \in P^{-1}(b_1)$, $\forall t \in I$. By uniqueness of path lifting, f, g have unique lifts to E, beginning at e_0. We also know that $\widetilde{F}(s,0)$ and $\widetilde{F}(s,1)$ are lifts of f, g both beginning at e_0.

Then $\widetilde{F}(s,0) = \widetilde{f}$ and $\widetilde{F}(s,1) = \widetilde{g}$ for all $s \in I$. Hence $\widetilde{F}(1,0) = \widetilde{F}(1,1) \implies \widetilde{f}(1) = \widetilde{g}(1)$, and $\widetilde{f} \simeq \widetilde{g}$. $\square$

Theorem 2.18. $\pi_1(S^1, (1,0) = b_0) \cong (\mathbb{Z}, +)$

Proof.

$$\begin{tikzcd} & \mathbb{R} \arrow[d, two heads, "p"] \\ I \arrow[ur, "\widetilde{f}"] \arrow[r, "f"'] & S^1 \end{tikzcd}$$

Consider the convering map

$$P : \mathbb{R} \to S^1,\ P(x) = (\cos(2\pi x), \sin(2\pi x))$$

Let $[f] \in \pi_1(S^1, b_0)$. Let $\widetilde{f}$ be the lift of f to $\mathbb{R}$ beginning at 0. Now consider $\widetilde{f}(1)$. It will be an integer since $\widetilde{f}(1) \in P^{-1}(b_0) = P^{-1}((1,0))$. Define

$$\phi : \pi_1(S^1, b_0) \to \mathbb{Z}, \quad \phi([f]) = f(1)$$

By previous lemma, ϕ is well-defined.

Now we show that ϕ is surjective. Let $n \in \mathbb{Z}$. Define $\widetilde{f}$ to be the path in $\mathbb{R}$ which starts at 0 and ends at n. Then by definition, $(P \circ \widetilde{f})$ will start at $(1,0)$ and end at $(1,0)$ where $\widetilde{f}$ is by definition the lift of $(P \circ \widetilde{f})$ beginning at 0. So $\phi([P \circ \widetilde{f}]) = n$.

To show that ϕ is injective suppose that $\phi([f]) = \phi([g]) = n$ for some $[f], [g] \in \pi_1(S^1, b_0)$ we need to show that $[f] = [g] \in \pi_1(S^1, b_0)$.

Now let $\widetilde{f}$ and $\widetilde{g}$ be two lifts of f and g respectively beginning at 0. Moreover, since $\mathbb{R}$ is simply connected and $\widetilde{f}(0) = \widetilde{g}(0) = 0$, $\widetilde{f}(1) = \widetilde{g}(1) = n$, $\widetilde{f} \simeq_p \widetilde{g}$, then $P \circ \widetilde{F}$ will be the path-homotopy between f and g. This implies $[f] = [g]$.

Now we want to show that ϕ is a group homomorphism. Let $[f], [g] \in \pi_1(S^1, b_0)$ such that $\phi([f]) = n$, $\phi([g]) = m$ for $n, m \in \mathbb{Z}$. Now we want to show that $\phi([f] \star [g]) = \phi([f \star g]) = n + m$. We need to find a lift of $f \star g$ beginning at 0 and ending at $n + m$.

Let $\widetilde{f}, \widetilde{g}$ be lifts of f, g respectively, $\widetilde{f}(0) = \widetilde{g}(0) = (0)$.

Define a path on $\mathbb{R}$ by

$$h(s) = \begin{cases} \widetilde{f}(2s), & s \in \left[0, \frac{1}{2}\right] \\ n + \widetilde{g}(2s - 1), & s \in \left[\frac{1}{2}, 1\right] \end{cases}$$

Then h is a path beginning at 0 and ending at $n+m$. Now consider,

$$P(h(s)) = \begin{cases} P \circ \widetilde{f}(2s), & s \in \left[0, \frac{1}{2}\right] \\ P \circ (n + \widetilde{g}(2s-1)) = P \circ \widetilde{g}(2s-1), & s \in \left[\frac{1}{2}, 1\right] \end{cases}$$

This implies that $h(s)$ is a lift of $f \star g(s) \implies \phi([f \star g]) = h(1) = n + m$ $\square$

Corollary 2.18.1. *Fundamental Theorem of Algebra.*

Proof. Let $p(z) = z^n + a_1 z^{n-1} + \cdots + a_n$, where $a_i \in \mathbb{C}$.
For the sake of contradiction assume that p has no roots in $\mathbb{C}$. Then for each $r \in \mathbb{R}^+$ define a function

$$f_r(s) = \frac{\frac{p(re^{2\pi i s})}{p(r)}}{\left|\frac{p(re^{2\pi i s})}{p(r)}\right|} \in \mathbb{C}$$

Note that $f_r(s)$ is a loop in S^1 based at $(1,0)$ for all r. ($f_r(0) = (1,0) = f_r(1)$ and $|f_r(s)| = 1, \forall s$). Note that as r varies f_r gives a path homotopy in S^1 between loops that are based at $(1,0)$. Now, note that $f_0(s)$ is the trivial loop based at $(1,0)$. This implies,

$$[f_r(s)] = [f_0(s)] = 0 \in \pi_1(S^1, (1,0))$$

Now fix a large value of r such that $r > |a_1| + \cdots |a_n|$ and $r > 1$. Now, we have for $|z| = r$

$$|a_1 z^{n-1} + \cdots + a_n| \leq |a_1| r^{n-1} + \cdots + |a_n|.$$

Thus if $r > |a_1| + \cdots + |a_n|$, then

$$|z^n| = r^n > r^{n-1}(|a_1| + \cdots + |a_n|) \geq |a_1 z^{n-1} + \cdots + a_n|.$$

Hence,

$$|z^n| > |a_1 z^{n-1} + \cdots + a_n|$$

This implies that the polynomial $p_t(z) = z^n + t(a_1 z^{n-1} + \cdots + a_n), \forall t \in [0,1]$ has no roots in circle $|z| = r$. If we now replace p by p_t in the definition of $f_r(s)$ we get

$$f_r(s,t) = \frac{\frac{p_t(re^{2\pi i s})}{p_t(r)}}{\left|\frac{p_t(re^{2\pi i s})}{p_t(r)}\right|}$$

At $t = 1$,

$$f_r(s,1) = f_r(s) \in \pi_1(S^1, (1,0))$$

At $t = 0$,

$$f_r(s,0) = e^{2\pi i n s}, \quad \forall s \in [0,1]$$

So $[f_r(s,0)] = n \in \mathbb{Z} \cong \pi_1(S^1, (1,0))$. But $f_r(s,t)$ as we vary $t \in [0,1]$ gives a path homotopy between $f_r(s,0)$ and $f_r(s,1) \implies [f_r(s,0)] = [f_r(s)] \in \pi_1(S^1, (1,0)) \implies n = 0$ contradicting that p is not constant. $\square$

Remark 2.19. *In general, we have $P : (E, e_0) \to (B, b_0)$ be a covering map. If E is path-connected then there exists a surjection $\phi : \pi_1(B, b_0) \to P^{-1}(b_0)$. If E is simply-connected, then ϕ is a bijection.*

Proof.

(a) For $[\gamma] \in \pi_1(B, b_0)$, let $\gamma' : I \to E$ be the unique lift of γ starting at e_0. Define
$$\phi : \pi_1(B, b_0) \to p^{-1}(b_0), \quad \phi([\gamma]) = \gamma'(1).$$
Note $\gamma'(1) \in p^{-1}(b_0)$ since $p(\gamma'(1)) = \gamma(1) = b_0$, so ϕ is well-defined.

To see surjectivity, let $e_\alpha \in p^{-1}(b_0)$. Since E is path-connected, there exists a path $\sigma_\alpha : I \to E$ such that $\sigma_\alpha(0) = e_0$ and $\sigma_\alpha(1) = e_\alpha$.

Then $\gamma = p \circ \sigma_\alpha$ is a loop in B based at b_0, since
$$\gamma(0) = (p \circ \sigma_\alpha)(0) = p(e_0) = b_0 \quad \text{and} \quad \gamma(1) = (p \circ \sigma_\alpha)(1) = p(e_\alpha) = b_0$$
By uniqueness of lifts, σ_α is the unique lift of γ starting at e_0, so $\phi([\gamma]) = \sigma_\alpha(1) = e_\alpha$. We can similarly define a unique lift for each e_α and thus ϕ is surjective.

(b) By part (a) we already know that ϕ is surjective. To see that ϕ is injective, let $[f], [g] \in \pi_1(B, b_0)$ such that $\phi([f]) = \phi([g]) = e_\alpha$ for some $e_\alpha \in p^{-1}(b_0)$, and let f', g' be the lifts of f, g starting at e_0.

Then $f'(0) = g'(0) = e_0$ and $f'(1) = g'(1) = e_\alpha$. Since E is simply connected, $\pi_1(E, e_0) \cong \{e\}$, i.e. any two paths with the same initial and final points in E are path-homotopic. Thus, $[f'] = [g']$, let F' be the homotopy between f' and g'. Then we have
$$F'(s, 0) = f'(s) \quad \text{and} \quad F'(s, 1) = g'(s).$$
Applying p to both sides,
$$(p \circ F')(s, 0) = p(f'(s)) = f(s) \implies p(f') = f$$
$$(p \circ F')(s, 1) = p(g'(s)) = g(s) \implies p(g') = g$$
Therefore $p \circ F' : I \times I \to B$ is a homotopy between f and g. Hence, $[f] = [g]$, ϕ is injective, and thus bijective.

$\square$

Theorem 2.20.
$$\pi_1(\mathbb{R}^2 \setminus \{(0,0)\}, b_0) \cong \mathbb{Z} \cong \pi_1(S^1), \quad b_0 \in S^1.$$

Proof. Let $x_0 \in S^1$. Then the inclusion map
$$J : (S^1, b_0) \to (\mathbb{R}^2 \setminus \{(0,0)\}, b_0)$$
induces an isomorphism
$$J_* : \pi_1(S^1, b_0) \to \pi_1(\mathbb{R}^2 \setminus \{(0,0)\}, b_0).$$
Consider
$$r : \mathbb{R}^2 \to S^1, \quad r(x) = \frac{x}{\|x\|}.$$
Observe that $r \circ J = \mathrm{id}$, thus
$$r_* \circ J_* = \mathrm{id}.$$
Thus J_* is injective.

Choose $[f] \in \pi_1(\mathbb{R}^2 \setminus \{(0,0)\}, b_0)$. Recall that,

$$J_* \circ r_*([f])$$

is explicitly $J \circ r \circ f$. Let

$$g(s) = \frac{f(s)}{\|f(s)\|}.$$

Next show $g \simeq_p f$. Define

$$F : I^2 \to \mathbb{R}^2 \setminus \{(0,0)\}$$

such that

$$F(s,t) = t \cdot \frac{f(s)}{\|f(s)\|} + (1-t)f(s).$$

F is the straight line homotopy and does not cross 0.
Since $f(s) \neq 0$ by definition, suppose

$$t \cdot \frac{1}{\|f(s)\|} + (1-t) = 0$$

implies $\|f(s)\|$ negative or $t \notin [0,1]$.
Thus F is well-defined, since $f \simeq_p g$,

$$J_* \circ r_*([f]) = [f].$$

Thus surjection. $\square$

3. Retractions

Definition 3.1. *If A is a subspace of X s.t. $\exists$ a map*

$$r : X \to A$$

with $r(a) = a$ for all $a \in A$, then we say A is a retract of X and r is a retraction map from X to A.

Definition 3.2. *Let A be a subspace of X. A is a deformation retract of X if there exists a homotopy*

$$H : X \times I \to X$$

such that

$$\begin{gathered} H(x,0) = x \quad \forall x \in X, \\ H(x,1) \in A, \\ H(a,t) = a. \end{gathered}$$

Theorem 3.3. *Let A be a deformation retract of X. Then $\iota : (A, x_0) \hookrightarrow (X, x_0)$ induces an isomorphism in $\pi_1(A, x_0)$ to $\pi_1(X, x_0)$. If A in a retract of X, then ι is injective.*

Proof.

(a) Let r be the map that retracts X to A, and $r(x_0) = a_0$ in A. Note that the composition $r \circ \iota$ is the identity on A. Now consider the homomorphism $(r \circ \iota)_\star$. We have,

$$(r \circ \iota)_\star = r_\star \circ \iota_\star = (\mathrm{id}_A)_\star = \mathrm{id}_{\pi_1(A,a_0)}$$

Let $\iota_\star([f]) = \iota_\star([g])$. Then,

$$\begin{aligned} & r_\star(\iota_\star([f])) = r_\star(\iota_\star[g]) \\ \implies & (\mathrm{id}_A)_\star([f]) = (\mathrm{id}_A)_\star([g]) \\ \implies & [f] = [g] \end{aligned}$$

Hence $\iota_\star$ is injective.

(b) Let H be the deformation retraction of X to A, and $H(x_0, 1) = a_0$ in A. From part (a) we know that $\iota_\star$ is injective. So we only need to show that $\iota_\star$ is surjective.
We define the continuous map $r = H(x,1)$. Consider the map $\iota \circ r$ which maps $X \to A \to X$. This map is homotopic to the identity map by H (by definition),

$$\iota \circ r \simeq \mathrm{id}_X$$

Then $(\iota \circ r)_\star = \iota_\star \circ r_\star \simeq (\mathrm{id}_X)_\star = \mathrm{id}_{\pi_1(X,x_0)}$ (Lemma 58.1). Since $\mathrm{im}(r_\star) = \pi_1(A, a_0)$, $\iota_\star$ is surjective. Thus, we have an isomorphism $\pi_1(A, a_0) \cong \pi_1(X, x_0)$.

$\square$

4. Applications of π_1-Computations

Theorem 4.1 (Brouwer-Fixed Point Theorem)**.** *Every map $h : D^2 \to D^2$ has a fixed point, i.e. $h(x) = x$ for at least one $x \in D^2$.*

Proof. Suppose towards contradiction $h(x) \neq x$ for all $x \in D^2$. Define a function $\gamma : D^2 \to S^1$ as follows: consider the straight line starting at $h(x)$ and going to x. Define $r(x)$ as the intersection of that line with $\delta D^2 = S^1$. Now r is a continuous function, and $r|_{S^1} = Id_{S^1}$. That is, r is a retraction of D^2 to S^1.

Since for the inclusion $\iota : S^1 \hookrightarrow D^2$ we should have $\pi_1(S^1, (1,0)) \xrightarrow{\iota_\star} \pi_1(D^2, (1,0))$ is an injection which gives a contradiction. $\square$

Theorem 4.2 (Borsuk-Ulam Theorem)**.** *Every map $f : S^2 \to \mathbb{R}^2$ has a pair of anti-podal points (x is an antipode of $-x$, $x \in S^2$) in S^2 such that $f(x) = f(-x)$.*

Proof. Suppose that this is false. Then $f(x) \neq f(-x)$ for all $x \in S^2$. We define $g : S^2 \to S^1$ by

$$g(x) = \frac{f(x) - f(-x)}{\|f(x) - f(-x)\|}$$

Define a loop η by the equator that lies in the $X - Y$ plane of S^2.

$$\eta(S) = (\cos(2\pi s), \sin(2\pi s), 0),\ s \in [0,1]$$

and let $h : I \to S^1$ defined by $h := g \circ \eta$. Note that since $g(x) = -g(-x)$ we have $h(s + \frac{1}{2}) = -h(s)$ for all $s \in \left[0, \frac{1}{2}\right]$.

$$\begin{aligned} g(\cos(2\pi(s+1/2)), \sin(2\pi(s+1/2)), 0) &= g(-\cos(2\pi s), -\sin(2\pi s), 0)) \\ &= -g(\cos(2\pi s), \sin(2\pi s), 0) \end{aligned}$$

By path-lifting lemma, we can lift h to a path in $\mathbb{R}$, say $\widetilde{h}$ starting at 0 such that

$$\widetilde{h}\left(s + \frac{1}{2}\right) = \widetilde{h}(s) + \frac{q(s)}{2}$$

for some odd integer values $q(s)$. But now, $q(s) = 2(\widetilde{h}\left(s + \frac{1}{2}\right) - \widetilde{h}(s))$ and hence it is also a continuous function. This implies $q(s)$ is a constant map (since the domain of q is conected), i.e. $\widetilde{h}\left(s + \frac{1}{2}\right) = \widetilde{h}(s) + \frac{q}{2}$.

$$\widetilde{h}(1) = \widetilde{h}\left(\frac{1}{2}\right) + \frac{q}{2} = \widetilde{h}(0) + q$$

This implies $\widetilde{h}(1) = q$ and hence $[h] \in \pi_1(S^1, (1,0))$ is not homotopic to the identity. But since η is nullhomotopic in S^2, $g \circ \eta$ is homotopic to a constant map $[h] = 0 \in \pi_1(S^1, (1,0))$. A contradiction! $\square$

5. Homotopy Equivalences

Theorem 5.1. *$\pi_1(X \times Y, x_0 \times y_0) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$ for any pair of topological spaces X and Y.*

Proof. Let $P_1 : X \times Y \to X$ and $P_2 : X \times Y \to Y$ be the respective projections.

$$\phi : \pi_1(X \times Y, x_0 \times y_0) \to \pi_1(X, x_0) \times \pi_1(Y, y_0)$$

where

$$\phi([f]) = (P_1)_\star[f] \times (P_2)_\star[f]$$

ϕ is surjective: Let $[g] \in \pi_1(X, x_0)$ and $[h] \in \pi_1(Y, y_0)$. Then define $[g] \times [h] : I \to X \times Y$ by $s \mapsto g(s) \times h(s)$. Then $\phi([g \times h]) = [g] \times [h]$.

ϕ is injective: Suppose that $[f] \in \pi_1(X \times Y, x_0 \times y_0)$ such that $\phi([f]) = (P_1)_\star([f]) \times (P_2)_\star([f]) = 0 = ([e_{x_0}] \times [e_{y_0}])$. Then $P_1 \circ f \simeq_p e_{x_0}$ and $P_2 \circ f \simeq_p e_{y_0}$ by homotopies F_t and G_t.

Then, $F_t \times G_t : I \to X \times Y$ gives $f \simeq_p e_{x_0 \times y_0} \implies [f] = 0$. $\square$

Definition 5.2. *We say $f : X \to Y$ is a homotopy equivalence if there exists $g : Y \to X$ such that $f \circ g \simeq id_Y$ and $g \circ f \simeq id_X$. Then we say that X is homotopy equivalent to Y and write it as $X \simeq_h Y$.*

Remark 5.3. *Homotopy equivalence is an equivalence relation on topological spaces.*

Proposition 5.4. *Suppose that X deformation retracts to A. Then $X \simeq_h A$.*

Proof. Let $H : X \times I \to A$ be the deformation retraction, and $\iota : A \hookrightarrow X$ be the inclusion map. Then let $r = H(x, 1)$ for all $x \in X$. Now note that,

$$r \circ \iota \simeq \mathrm{id}_A$$

And, by definition of the deformation retraction H gives us,

$$\iota \circ r \simeq \mathrm{id}_X$$

Thus, $X \simeq_h A$. $\square$

Remark 5.5. *Note not all homotopy equivalence come from deformation retractions.*

Remark 5.6. *π_1 is not only an invariant of homeomorphism class of a topological space but also an invariant of the homotopy equivalence class of the space.*

Lemma 5.7. *If $\phi_t : X \to Y$ be a homotopy and h is the path given by $\phi_t(x_0)$ for a choice of $x_0 \in X$, then we have $(\phi_0)_\star = \hat{\hat{h}}((\phi_1)_\star)$ on $\pi_1(X, x_0)$. That is,*

$$\begin{tikzcd}
\pi_1(X, x_0) \arrow[r, "(\phi_1)_\star"] \arrow[dr, "(\phi_0)_\star"'] & \pi_1(Y, \phi_1(x_0)) \arrow[d, "\hat{\hat{h}}"] \\
& \pi_1(Y, \phi_0(x_0))
\end{tikzcd}$$

Proof. Let $[f] \in \pi_1(X, x_0)$. Let h_t be a reparameterization of the path h such that $h_t(s) := h(st)$. In particular, h_t is a path from $\phi_0(x_0)$ to $\phi_t(x_0)$. Now the composition, $h_t \star \phi_t \circ f \star \overline{h}_t$ gives a path homotopy between

$$h_0 \star \phi_0 \circ f \star \overline{h}_0 = e_{x_0} \star \phi_0 \circ f \star \overline{e}_{x_0} \simeq \phi_0 \circ f$$

and

$$h_1 \star \phi_1 \circ f \star \overline{h}_1 = h \star \phi_0 \circ f \star \overline{h} \implies (\phi_0)_\star = \widehat{\overline{h}}((\phi_1)_\star)$$

□

Theorem 5.8. *Let $\phi : X \to Y$ be a homotopy equivalence. Then $\phi_\star : \pi_1(X, x_0) \to \pi_1(Y, \phi(x_0))$ is an isomorphism.*

Proof. Since ϕ is a homotopic equivalence, there exists a $\psi : Y \to X$ such that $\phi \circ \psi \simeq \mathrm{id}_Y$ and $\psi \circ \phi \simeq \mathrm{id}_X$. Consider,

$$\pi_1(X, x_0) \xrightarrow{\phi_\star} \pi_1(Y, \phi(x_0)) \xrightarrow{\psi_\star} \pi_1(X, \psi(\phi(x_0))) \xrightarrow{\phi_\star} \pi_1(Y, \phi(\psi(\phi(x_0))))$$

Note that $\psi_\star \circ \phi_\star = (\psi \circ \phi)_\star = \widehat{\overline{h}} \circ (\mathrm{id}_X)_\star$ for some path h in X (by the previous lemma). Tha implies $\psi_\star \circ \phi_\star = \widehat{\overline{h}} \implies \psi_\star$ is surjective since $\widehat{\overline{h}}$ is an isomorphism.

$\psi_\star$ is injective because

$$\phi_\star \circ \psi_\star = \widehat{\overline{h'}} \circ (\mathrm{id}_Y)_\star$$

for some path h' in Y (by the previous lemma).

Hence, $\phi_\star = (\psi_\star)^{-1} \circ \widehat{\overline{h}}$ which is an isomorphism. □

6. $\pi_1(S_n)$

Lemma 6.1. *If a space X is the union of a collection of path-connected open sets A_α each containing the base point $x_0 \in X$. Let $A_\alpha \cap A_\beta$ be path connected for any α, β, then every loop in X based at x_0 is path homotopic to a composition of loops each contained in a single A_α for some α and based at x_0.*

Proof. Let $[f] \in \pi_1(X, x_0)$. Suppose that $0 = S_0 < S_1 < \ldots < S_m = 1$ is a subdivision of $[0,1]$ such that each interval $[S_{i-1}, S_i]$ is mapped by f to a single A_α, let f_i be the path obtained by the restriction $f|_{[S_{i-1},S_i]}$. Let us denote that A_α by A_i. Note $f(0) = x_0$. Define g_i to be the path from x_0 to $f(S_i) \in A_i \cap A_{i+1}$.

Consider,

$$(f_1 \star \overline{g_1}) \star (g_1 \star f_2 \star \overline{g_2}) \star (g_2 \star f_3 \star \overline{g_3}) \star \cdots \star (g_{m-1} \star f_m) \simeq_p f$$

while each of them are loops based at x_0 in A_is as i varies. □

Theorem 6.2. $\pi_1(S^n, x_0) \cong \{0\}$ *if* $n \geq 2$.

Proof. On S^2 consider two open sets $A_1 = S^2 \setminus \{\text{north pole}\}$ and $A_2 = S^2 \setminus \{\text{south pole}\}$. Let $[f] \in \pi_1(S^2, x_0)$. Then by the lemma $f \simeq_p g_1 \star g_2$ where $[g_1] \in \pi_1(A_1, x_0)$ and $[g_2] \in \pi_1(A_2, x_0)$ for some g_1 and g_2 loops based at x_0. But since $\pi_1(A_1) = \pi_1(A_2) = 0$,

$$[f] = 0 \in \pi_1(S^2, x_0) \implies \pi_1(S^2, x_0) = 0$$

(Same proof for any $n \geq 2$). □

Remark 6.3 (Stereographic Projection). *$S^n \setminus \{$ some pt $\}$ is homeomorphic to $\mathbb{R}^n$.*

7. Covering Spaces (again)

Remark 7.1. *For every group G there exists a space X_G such that $\pi_1(X_G) \cong G$.*

Definition 7.2. *$\mathbb{RP}^n$ can be defined the following ways:*

(1) The quotient of $\mathbb{R}^{n+1} \setminus \{0\}$ under the equivalence relation $x \sim \lambda x$ for $0 \neq \lambda \in \mathbb{R}$.
(2) The quotient of S^n under the equivalence relation $x \sim -x$.

Theorem 7.3. *Define $p : S^n \to \mathbb{RP}^n$, $x \mapsto [x]$. Then p is a covering map.*

Proof. First show that p is an open map (i.e. U open $\implies$ $p(U)$ open). Let $V \subset S^n$ be an open set (S^n has subspace topology from $\mathbb{R}^{n+1}$). Consider 'antipodal map',

$$a : S^n \to S^n, \quad x \mapsto -x$$

Then a is a homeomorphism. Thus its inverse is continuous, and $a(V)$ is open in S^n. Now since $p^{-1}(p(V)) = V \cup a(V)$, which is open in S^n, p is an open map (using definition of quotient topology).

Now we will show p is a covering map. Let $y \in \mathbb{R}P^n$. Pick $x \in p^{-1}(y)$. Choose a small neighborhood U of x in S^n such that U contains no antipodal points. Then $p|_U : U \to p(U)$ is a homeomorphism (since p is open and continuous and a bijection).

Similarly, $p|_{a(U)} : a(U) \to p(a(U))$ is also an homeomorphism. Note $p(a(U)) = p(U)$ which is an open set around $y \in \mathbb{RP}^n$. So each point in $\mathbb{RP}^n$ is evenly covered by an open set in $\mathbb{RP}^n$. Hence, p is a covering map (specifically a 2-fold covering).

Then $\pi_1(\mathbb{RP}^n, x_0) \leftrightarrow p^{-1}(x_0) \implies |\pi_1(\mathbb{RP}^n, x_0)| = 2 \implies |\pi_1(\mathbb{RP}^n, x_0)| = \mathbb{Z}/2\mathbb{Z}$. ☐

Corollary 7.3.1. *$\mathbb{RP}^n$ is not homotopy equivalent to S^n for $n \geq 2$.*

Proof. Fundamental group is a homotopy-equivalence invariant, and S^n for $n \geq 2$ is simply connected. ☐

Remark 7.4. *Homeomorphism $X \cong Y \implies X \simeq_h Y$ (homotopy equivalence).*

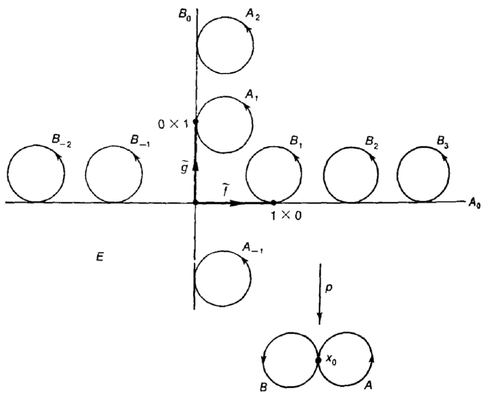
Theorem 7.5. *$\mathbb{R}^2$ is not homeomorphic to $\mathbb{R}^n$ for $n \neq 2$.*

Proof. Suppose for contradiction there exists a homeomorphism f from $\mathbb{R}^2 \to \mathbb{R}^n$. Then $f\ |_{\mathbb{R}^2 \setminus \{pt\}}$: $\mathbb{R}^2 \setminus \langle x \rangle \to \mathbb{R}^n \setminus \{f(x)\}$ is also a homeomorphism. So $\mathbb{R}^2 \setminus \{x\} \simeq_h \mathbb{R}^n \setminus \{f(x)\}$ must be homotopy equivalent, But π_1 is a homotopy equivalence so we should have $\pi_1(\mathbb{R}^2 \setminus \{x\}) \cong \pi_1(S^1) \cong \pi_1(\mathbb{R}^n \setminus \{f(x)\}) \implies \mathbb{Z} \cong 0$. A contradiction! ☐

Theorem 7.6. *Fundamental group of the 'figure eight space' is not abelian.*

Proof. Let X be the figure eight space. Then $X = S^1 \sqcup S^1/(1,0) \sim (1,0)$ with the quotient space topology. Let x_0 be the base point. Define a covering space $E \subseteq \mathbb{R}^2$, where

$$E = \{\text{x-axis}\} \cup \{\text{y-axis}\} \cup \{\text{a circle tangent to both axes at each nonzero integer point}\}$$



We claim that there exists a projection map $p : E \to X$. Consider two paths $\widetilde{f}, \widetilde{g} : I \to E$ defined by $\widetilde{f}(s) = (s, 0)$, $\widetilde{g}(s) = (0, s)$. Define $f := p \circ \widetilde{f}$, $g := p \circ \widetilde{g}$. That is f wraps once around A, g wraps once around B.

Claim: $[f \star g] \neq [g \star f] \in \pi_1(\infty, x_0)$.

Proof: Claim $\iff$ $f \star g \not\simeq_p g \star f$. For the sake of contradiction suppose $f \star g \simeq_p g \star f$. Now consider the lift of $f \star g$ beginning at $(0, 0)$ to E given by $\widetilde{f} \star B_1$, and the lift of $g \star f$ beginning at $(0, 0)$ to E given by $\widetilde{g} \star A_1$. If they are path-homotopic, they should have the same end points, but $(\widetilde{f} \star B_1)(1) = (1, 0) \neq (0, 1) = (\widetilde{g} \star A_1)(1)$. $\square$

Corollary 7.6.1. *$\pi^2 \# \pi^2 \not\cong S^2, \mathbb{RP}^2$, where π^2 is the hollow torus, and # is the connected sum.* ***TODO: ??***

Proposition 7.7. *Let $h : S^1 \to Y$ where Y is some topological space, then h is homotopic to a constant map iff h can be extended to a continuous map $D^2 \to Y$.*

Proof. ($\implies$)

Let $H : S^1 \times I \to Y$ be a homotopy from h to a constant map $S^1 \to x_0$. Define $\pi : S^1 \times I \to D^2$ by $\pi(x, t) = (1 - t)x$. Then $\pi \mid_{S^1 \times [0,1)}$ is a bijection to $D^2 \setminus \{0\}$ and it maps $S^1 \times \{1\}$ to $0 \in D^2$.

Since $S \times I$ is compact and D^2 is Hausdorff, π is a closed map. Now H induces a map $g : D^2 \to Y$ such that $g \circ \pi = H$, given by

$$g(x) = \begin{cases} H(\pi^{-1}(x)) & x \neq 0 \\ y_0 & x = 0 \end{cases}$$

The map g is continuous since π is a quotinent map. Thus the map g is the desired extension of h for if $x \in S^1$ then,

$$g(x) = g(\pi(x, 0)) = H(x, 0) = h(x)$$

($\impliedby$)

Let $g : D^2 \to Y$ be a continuous extension of h, and define $F : S^1 \times I \to Y$ by

$$F(x, t) = g((1 - t)x)$$

Then F is a homotopy between h and a constant map. $\square$

Remark 7.8. *By the Galois correspondence, covers of X are in bijection with subgroups of $\pi_1(X)$.*

Proposition 7.9. *Let $p : (\widetilde{X}, \widetilde{x}_0) \to (X, x_0)$ be a covering map. Then the map p_* is injective. Moreover, $p_*(\pi_1(\widetilde{X}, \widetilde{x}_0))$ consists of homotopy classes of loops in X that lift to loops in $\widetilde{X}$ based at $\widetilde{x}_0$.*

Proof. Let $[f] \in \pi_1(\widetilde{X}, \widetilde{x}_0)$ and suppose $p_*([f]) = 0 \in \pi_1(X, x_0)$. Then there exists a path homotopy H between $p \circ f$ and the constant loop c_{x_0}. By the homotopy lifting lemma, we can lift this to a homotopy between f and the trivial loop $c_{\widetilde{x}_0}$ in $\widetilde{X_0}$. Thus, $[f] \simeq_p 0$, and p_* is injective.

Moreover, loops at x_0 lifting to loops at $\widetilde{x}_0$ certainly represent elements of $p_*(\pi_1(\widetilde{X}, \widetilde{x}_0))$, and conversely, a loop representing an element of $p_*(\pi_1(\widetilde{X}, \widetilde{x}_0))$ is homotopic to a loop having sucha lift, so by homotopy lifting, the loop itself must have such a lift. $\square$

Definition 7.10. *Given $p : \widetilde{X} \to X$ a covering, $|p^{-1}(x)|$ for $x \in X$ is the number of **sheets**.*

Proposition 7.11. *Given X and $\widetilde{X}$ path connected, then the number of sheets of $p : (\widetilde{X}, \widetilde{x}_0) \to (X, x_0)$ of the covering equals the index of $p_*(\pi_1(\widetilde{X}, \widetilde{x}_0))$ in $\pi_1(X, x_0)$.*

Proof. Let g be a loop based at x_0 in X. Let $\tilde{g}$ be its lift in $\widetilde{X}$, starting at $\widetilde{x}_0$.

Now let $[h] \in H := p_*(\pi_1(\widetilde{X}, \widetilde{x}_0))$, and let $\widetilde{h} \circ \widetilde{g}$ be a lift based at $\widetilde{x}_0$, starting at $\widetilde{x}_0$. We look at the lift of $h \circ g$, $\widetilde{h} \circ \widetilde{g}$ which ends at the same point as $\widetilde{g}$ since $\widetilde{h}$ is a loop at $\widetilde{x}_0$.

Thus we can define a function ϕ from the cosets $H[g]$ to $p^{-1}(x_0)$. ϕ is surjective since $\widetilde{X}$ is path-connected and hence $\widetilde{x}_0$ can be joined to any point in $p^{-1}(x_0)$ by a path $\widetilde{g}$ that projects to a loop g at x_0.

To see that ϕ is injective. Let $\phi(H[g_1]) = \phi(H[g_2])$. Then, the lifts correspoding to g_1 and g_2 end at the same point, and thus $[g_1 \circ \overline{g_2}] = [g_1][g_2]^{-1} \in H \implies H[g_1] = H[g_2]$.

Thus there exists a bijection between the set of cosets of $\pi_1(X, x_0)$ and the fibre $p^{-1}(x_0)$ and $|p^{-1}(x_0)| =$ number of cosets $= [\pi_1(X, x_0) : H]$. $\square$

Proposition 7.12. *Given a covering space $p : (\widetilde{X}, \widetilde{x}_0) \to (X, x_0)$ and a map $f : (Y, y_0) \to (X, x_0)$ with $p(\widetilde{x}_0) = x_0$ and $f(y_0) = y_0$ with Y path-connected and locally path-connected, then there exists a lift $\widetilde{f} : (Y, y_0) \to (\widetilde{X}, \widetilde{x}_0)$ of f iff $f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\widetilde{X}, \widetilde{x}_0))$.*

Proof. ($\implies$)

Assume there exists a lift $\widetilde{f}$ then we know that $f_* = p_* \circ \widetilde{f}_*$, and thus

$$\operatorname{im}(f_*) = \operatorname{im}(p_* \circ \widetilde{f}_*) \subseteq \operatorname{im}(p_*) \implies f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\widetilde{X}, \widetilde{x}_0))$$

($\impliedby$)

Assume $f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\widetilde{X}, \widetilde{x}_0))$. Let $y \in Y$ and γ a path in Y from y_0 to y. Let $\widetilde{f\gamma}$ be the unique lift of the path $f \circ \gamma$ in X starting at $\widetilde{x}_0$. Now define $\widetilde{f} : (Y, y_0) \to (\widetilde{X}, \widetilde{x}_0)$

$$\widetilde{f}(y) = \widetilde{f\gamma}(1)$$

This is well-defined, let δ be some arbitrary path from y_0 to y. Then $f_\gamma \cdot \overline{f_\delta}$ is a loop based at x_0.

$$[f_\gamma] \cdot [f_\delta]^{-1} \in f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\widetilde{X}, \widetilde{x}_0)), \quad \text{by hypothesis.}$$

Thus there exists a path homotopy from $f\gamma$ to $f\delta$ to a loop in X which lifts to a loop in $\widetilde{X}$, based at $\widetilde{x}_0$. **TODO:** details idrc

Now we prove continuity of $\widetilde{f}$. Let $y \in Y$. Choose $U \subseteq X$ that is an openly covered neighbourhood of $f(y)$ having a lift $\widetilde{U}$ such that $p : \widetilde{U} \to U$ is a homeomorphism. Also pick a path-connected open neighbourhood V of y s.t. $f(V) \subseteq U$.

Choose $y' \in V$. Let δ be a path from y_0 to y and η a path from y to y' in V. Thus $f\delta \cdot f\eta$ lifts to $\widetilde{f\delta} \cdot \widetilde{f\eta}$, where $\widetilde{f\eta} = p^{-1} f\eta$ and $p^{-1} : U \to \widetilde{U}$. Thus, $\widetilde{f}(V) \subset \widetilde{U}$ and $\widetilde{f}\mid_V = p^{-1} f$. Hence $\widetilde{f}$ is continuous at y. $\square$

8. Classification of Covering Space

Definition 8.1. *We say that a covering space $\widetilde{X}$ of X is a universal cover if $\pi_1(\widetilde{X}) = 0$.*

Example 8.1. *(1) $\mathbb{R}$ is a universal cover of S^1.*
(2) Every simply connected space is a universal cover of itself.
(3) $\mathbb{C} \to \mathbb{C}$ where $z \mapsto z^2$.
(4) S^n is a universal cover of $\mathbb{RP}^n$ for $n \geq 2$

Remark 8.2. *An object with property x is universal if any other object with property x factors through the first object.*

Theorem 8.3. *Suppose that $\widetilde{X}$ is a universal cover of X, then if X is path-connected and locally-connected, any other path-connected cover X_1 of X, $\widetilde{X}$ covers X_1. Moreover the following diagram commutes.*

$$\begin{tikzcd} \widetilde{X} \arrow[r, dashed, "p_1"] \arrow[rr, bend right, "p"'] & X_1 \arrow[r, "p_2"] & X \end{tikzcd}$$

Proof. The proof follows from the following two lemmas. □

Lemma 8.4. *Suppose that $X, X_1, \widetilde{X}$ are all path-connected such that $X_1 \xrightarrow{p_2} X$, $\widetilde{X} \xrightarrow{p} X$ are covers with $\pi_1(\widetilde{X}) = 0$ then $\exists\, p_1 : \widetilde{X} \to X_1$ such that $p_1 \circ p_2 = p$.*

Proof. Follows from the general lifting proposition. □

Lemma 8.5. *Let $X, X_1, \widetilde{X}, p, p_1, p_2$ be defined as before, then p_1 is a covering map.*

Proof. Let $x_1 \in X_1$ be a point, U be a neighborhood of $p_2(x_1) \in X$ that is evenly covered by both p_2 and p. Now let $p^{-1}(U) = \sqcup_\alpha U_\alpha$ and $p_2^{-1}(U) = \sqcup_{\alpha'} V_{\alpha'}$. Moreover assume that V is in the slice of $p_2^{-1}(U)$ where x_1 lies.

Claim: V is evenly covered by p_1.

Proof: Consider the U_α's in $\widetilde{X}$ that are mapped to V. Lets say that collection is $\{U_\beta\}$. Then $p_1^{-1}(V) = \sqcup_\beta U_\beta$ and $p_1|_{U_\beta}$ is a homeomorphism since it is a composition of $p|_{U_\beta}$ and $(p_2|_V)^{-1}$ which are both homeomorphisms □

Theorem 8.6. *Let $\widetilde{X}$ be a universal cover of X, then $\widetilde{X}$ is unique upto homeomorphism.*

Proof. Hint: Lifts are unique if we fix a base point of the lift.

Let $\widetilde{X}$ and $\overline{X}$ be two universal covers of X. Then,

$$\begin{tikzcd} & (\overline{X}, \overline{x_0}) \arrow[d, "p_2"] \\ (\widetilde{X}, \widetilde{x_0}) \arrow[ur, "f"] \arrow[r, "p_1"] & (X, x_0) \end{tikzcd}.$$

$$\begin{tikzcd}
 & (\widetilde{X}, \widetilde{x_0}) \arrow[d, "p_1"] \\
(\overline{X}, \overline{x_0}) \arrow[ur, "g"] \arrow[r, "p_2"] & (X, x_0)
\end{tikzcd} .$$

$$p_1 = p_2 \circ f, \qquad p_2 = p_1 \circ g$$

That is,

$$p_1 = p_1 \circ g \circ f$$

Moreover, $g \circ f(\widetilde{x_0}) = \widetilde{x_0}$. Hence, $g \circ f$ is a lift of p_1 to $\widetilde{X}$ by p_1. Identity is also such a lift, then $g \circ f = \text{id}$ (by uniqueness of lift homotopy). $\square$

TODO: Read this proof ?? For any sub space of $\pi_1(X) \leftrightarrow$ cover X_1 of X with $p_\star(\pi_1(X_1)) \cong$ that subspace.

Remark 8.7. *Any collection of homomorphisms $\phi_\alpha : G_\alpha \to H$ extends uniquely to $\phi : \star_\alpha G_\alpha \to H$ such that $\phi(g_1 \cdots g_n) \mapsto \phi_{\alpha_1}(g_1)\phi_{\alpha_2}(g_2) \ldots \phi_{\alpha_n}(g_n)$ where G_α is a free group, H is a group, and $\star_\alpha G_\alpha$ consists of all words of arbitrary length where each letter belongs to a G_{α_i} and consequetive letters are from different G_{α_i}.*

Now suppose that X is decomposed as a union of path-connected spaces $\{A_\alpha\}$ each of which contains the base point $x_0 \in X$. Then inclusion $A_\alpha \hookrightarrow X \implies j_\alpha : \pi_1(A_\alpha) \to \pi_1(X)$ induce $\Phi : \star_\alpha \pi_1(A_\alpha) \to \pi_1(X)$.

Let $(\iota_{\alpha\beta})_\star : \pi_1(A_\alpha \cap A_\beta) \to \pi_1(A_\alpha)$ be induced from inclusion. Hence,

$$(j_\alpha)_\star \circ (\iota_{\alpha\beta})_\star = (j_\beta)_\star \circ (\iota_{\beta\alpha})_\star : \pi_1(A_\alpha \cap A_\beta) \to \pi_1(X)$$

If $w \in \pi_1(A_\alpha \cap A_\beta)$ then $(\iota_{\alpha\beta})_\star(w) \left((\iota_{\beta\alpha})_\star(w)\right)^{-1}$ is in the kernel of Φ.

Let N denote the normal subgroup of $\star_\alpha \pi_1(A_\alpha)$ generated by such elements. Then $\ker(\Phi) = N$ *(under assumptions).*

Theorem 8.8 (Seifert-Van Kampen Theorem)**.** *If X is a union of path-connected open sets A_α such that $x_0 \in X$, and $x_0 \in A_\alpha$ for all α, and $A_\alpha \cap A_\beta$ is path-connected, and $A_\alpha \cap A_\beta \cap A_\gamma$ is also path-connected for all α, β, γ. Then Φ induces an isomorphism,*

$$\pi_1(X) \cong \star_\alpha \pi_1(A_\alpha)/N$$

Proof. We have shown before that Φ is surjective. Now we will show that $\ker(\Phi) = N$. Define a factorization of an element $[f] \in \pi_1(X)$ as the product $[f_1] \ldots [f_k]$ such that each $[f_i] \in \pi_1(A_\alpha)$ for some A_α and $f \sim_p f_1 \star f_2 \star \cdots \star f_k$. A factorization of f is thus a word in $\star_\alpha \pi_1(A_\alpha)$. We will say that any two such factorizations are equivalent if they are related by the following two moves or their inverses:

(1) Combine adjacent terms $[f_i][f_{i+1}]$ into a single term if $[f_i][f_{i+1}]$ are in the same group.
(2) If $f_i \in A_\alpha \cap A_\beta$ we may regard $[f_i] \in \pi_1(A_\alpha)$ as $[f_i] \in \pi_1(A_\beta)$.

The first move does not change the factorization. The second move does not the element in $\star_\alpha \pi_1(A_\alpha)/N$.

Let us assume that $[f_1]\dots[f_k]$ and $[f_1']\dots[f_l']$ are two factorizations of $[f] \in \pi_1(X)$. We will show that they are equivalent. We have,

$$f_1 \star \cdots \star f_k \cong f_1' \star \cdots \star f_l'$$

That is, there exists a path-homotopy $F : I \times I \to X$. Now choose a partition of $I \times I$ by

$$0 = s_0 < \cdots < s_m = 1 \text{ and } 0 = t_0 < \cdots < t_n = 1$$

such that each rectangle $R_{ij} = [s_{i-1}, s_i] \times [t_{j-1}, t_j]$ is mapped to a single A_α by F. We label this A_α by A_{ij}.

Now we will modify the partition in two ways:

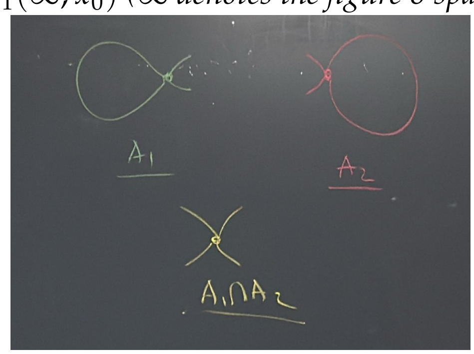
(1) Add subdivisions to the top ($[t_{n-1}, t_n]$) and bottom row ($[t_0 - t_1]$) such that for each end point in the partition there exists a corresponding s_i.
(2) Have at least 3 rows.
(3) Every point in $I \times I$ must lie in at most 3 (smaller) rectangles
(4) Rename the smaller rectangles (left to right, bottom to top) using R_i.

Define γ_r as the path from the left to the right edge of $I \times I$, and it separates rectangles $R_1, \dots R_r$ from the rest of the rectangles.

Note that F_{γ_r} is a loop based at x_0. And all F_{γ_r} are path homotopic. Then $F|_{\gamma_r}$ has a factorization. If $F(v) \neq x_0$, then we have a path from $x_0 \to F(v) = F|_{\gamma_1}$ $\square$

Example 8.2.

(1) $\pi_1(\infty, x_0)$ *(∞ denotes the figure 8 space).*



Then, $\pi_1(\infty, x_0) \cong (\pi(A_1) \times \pi_1(A_2))/N = (\mathbb{Z} \times \mathbb{Z})/N$. *Since* $N = A_1 \cap A_2$ *deformation retracts to a point this is* $\mathbb{Z} \times \mathbb{Z}$.

(2) $\pi_1(S^1 \vee S^2) \cong \mathbb{Z}$
$X_1 \vee X_2 := (X_1 \sqcup X_2)/(x_1 \sim x_2)$ *for some* $x_1 \in X_1$, $x_2 \in X_2$. *Then,* $\pi_1(S^1 \vee S^2) = \pi_1(S^1) \times \pi_1(S^2)/\pi_1(\{x_0\}) = \mathbb{Z}$.

(3) $\pi_1(\mathbb{R}^3 \setminus S^1) = \pi_1(S^1 \vee S^2) = \mathbb{Z}$.

9. Cell Complexes

Definition 9.1. *We start with a discrete space whose points are regarded as 0-cells. Now, we inductively define the n-**skeleton** X^n from X^{n-1} by attaching n-cells e^n_α via maps $\phi_\alpha : S^{n-1} \to X^{n-1}$ where e^n_αs are D^n_α. Hence, $X^n = X^{n-1} \sqcup D^n_\alpha/(x \sim \phi_\alpha(x) \text{ if } \alpha, \forall x \in S^{n-1})$. We say a topological space X has a CW/CX structure if $X = \cup_n X^n$.*

Definition 9.2. *Each cell e^n_α of a CW CX has a map $\Phi_\alpha : D^n_\alpha \to X$ which extends $\phi_\alpha : \partial D^n_\alpha \to X$. It is defined by*

$$D^n_\alpha \hookrightarrow X^{n-1} \sqcup D^n_\alpha \to \underset{\phi_\alpha : \partial D^n_\alpha \to X^{n-1}}{X^{n-1} \sqcup D^n_\alpha} = X^n \hookrightarrow x$$

*This map is called the **characteristic map** asssociated to ϕ_α.*

Definition 9.3. *A **subcomplex** A of a CW CX X is a subspace $A \subset X$ such that A is a union of some number of cells in X with a attaching map for each of those cells has a image contained in A. We say that (X, A) is a CW pair.*

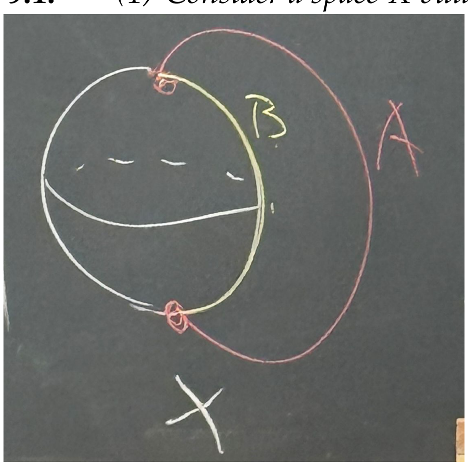
Definition 9.4 (Quotinent CX)**.** *If (X, A) is a CW pair, then X/A inherits a canonical cell structure from X. The cells of X/A are the cells $X - A$, plus one 0-cell given by the image of A in X/A. For a cell e^n_α of $X - A$ with $\phi_\alpha : S^{n-1} \to X^{n-1}$, the corresponding attaching map in X/A is*

$$S^{n-1} \to X^{n-1} \to X^{n-1}/A^{n-1}$$

Theorem 9.5. *If (X, A) is a CW pair and A is contractible then $q : X \to X/A$ is a homotopy equivalence.*

Proof. Hatcher Chapter 0, Homotopy Extension Lemma ☐

Example 9.1. *(1) Consider a space X built from S^2 by attaching an arc to two points in S^2.*



Both (X, A) and (X, B) are CW complexes.

(2) $\pi_1(S^3, S^1) = \pi_1(\mathbb{R}^3, \{line\}) = \pi_1(R^2, \{pt\}) = \mathbb{Z}$

10. Homology

Remark 10.1. *$\pi_n(S^n) = \mathbb{Z}$, and $\pi_m(S^n) = 0$ if $m < n$. We don't know how to compute $\pi_m(S^n)$ for $m > n$.*

Homology distinguishes higher dimension space and it is easy to compute. We define simplical and singular homology separately, but eventually they have the same properties.

Remark 10.2. *For spheres the homology groups $H_i(S^n)$ are isomorphic to the homotopy groups $\pi_i(S^n)$ for $1 \le i \le n$; and for $i > n$, $H_i(S^n) = 0$.*

10.1. **Simplical Homology.**

Definition 10.3. *A **n-simplex** is the smallest convex set in $\mathbb{R}^m$ spanned by $(n+1)$ points $v_0, \dots, v_n$ such that all of them do not lie in any hyperplane of dimension less than n. That is, $v_1 - v_0$, $v_2 - v_0, \dots, v_n - v_0$ are linearly independent. We say $(v_0, \dots, v_n)$ comes with the order $v_0, \dots, v_n$.*

The standard n-simplex is

$$\left\{ (t_0, \dots, t_n) \in \mathbb{R}^{n+1} \mid \sum_i t_i = 1, t_i \ge 0 \right\}$$

There is a canonical homeomorphism from Δ^n to any n-simplex.

$$(t_0, \dots, t_n) \mapsto \sum t_i v_i$$

Definition 10.4. *A **face** of an n-simplex $\Delta^n = [v_0, \dots, v_n]$ is the $n-1$-simplex obtained by deleting one of its $n+1$ vertices. The vertices of a face, or of any subsimplex spanned by a subset of the vertices, will always be ordered according to their order in the larger simplex.*

Definition 10.5. *The **boundary** $\partial\Delta^n$ is the union of all the faces of Δ^n.*

Definition 10.6. *The **open simplex** $\mathring{\Delta}^n$ is the interior of Δ^n, i.e. $\Delta^n - \partial\Delta^n$.*

Definition 10.7. *The **Δ-complex structure** of a space X is a collection of maps $\sigma_\alpha : \Delta^n \to X$.*

(1) $\sigma_\alpha|_{\mathring{\Delta}^n}$ is injective and each point of X is in the image of exactly one such $\sigma_\alpha|_{\mathring{\Delta}^n}$.
(2) Restriction of σ_α to a face of Δ^n is one of the maps $\sigma_\beta : \Delta^{n-1} \to X$. We identify the face of Δ^n with Δ^{n-1} by the canonical linear homeomorphism between them that preserves the ordering of the vertices.

$$\lambda_i : \Delta^{n-1} \to \Delta^n, \qquad (t_0, \dots, t_{n-1}) \mapsto (t_0, \dots, t_{i-1}, 0, t_i, \dots, t_{n-1})$$

(3) A set $U \subset X$ is open iff $\sigma_\alpha^{-1}(U)$ is open in Δ^n, for all α.

Definition 10.8. *Let X be a Δ-complex and let $\Delta_n(X)$ denote the free abelian group with basis n-simplicies of X. The elements of $\Delta_n(X)$ are of the form $\sum_i n_i\sigma_i$ where this is a finite sum, $n_i \in \mathbb{Z}$ and $\sigma_i : \Delta^n \to X$.*

We denote the deletion of v_i from the simplex as $[v_0, \dots, \widehat{v_i}, \dots, v_n] \cong [v_0, \dots, v_{n-1}]$. We will always orient the 1-simplex $[v_i, v_j]$ by an arrow going from v_i to v_j for $i < j$.

Definition 10.9. *We define a **boundary homomorphism** $\partial_n : \Delta_n(X) \to \Delta_{n-1}(X)$ where,*

$$\partial_n(\sigma_\alpha) = \sum_i (-1)^i \sigma_\alpha \mid_{[v_0,\ldots,\widehat{v}_i,\ldots,v_n]}$$

Proposition 10.10. *The composition*

$$\Delta_n(X) \xrightarrow{\partial_n} \Delta_{n-1}(X) \xrightarrow{\partial_{n-1}} \Delta_{n-2}(X)$$

is zero ($\partial_{n-1} \circ \partial_n = 0$).

Proof.

$$\partial_n(\sigma) = \sum_i (-1)^i \sigma|_{[v_0,\ldots,\widehat{v}_i,\ldots,v_n]}$$

$$\partial_{n-1} \circ \partial_n(\sigma) = \sum_{j<i} (-1)^i (-1)^j \sigma|_{[v_0,\ldots,\widehat{v}_j,\ldots\widehat{v}_i,\ldots,v_n]} + \sum_{i<j} (-1)^i (-1)^{j-1} \sigma|_{[v_0,\ldots,\widehat{v}_i,\ldots\widehat{v}_j,\ldots,v_n]}$$

□

Definition 10.11. *In general, whenever we have (C_n, ∂_n)*

$$\cdots \to C_{n+1} \xrightarrow{\partial_{n+1}} C_n \xrightarrow{\partial_n} C_{n-1} \xrightarrow{\partial_{n-1}\cdots}$$

*with $\partial_n \circ \partial_{n+1} = 0$, then $\operatorname{img}(\partial_{n+1}) \subset \ker(\partial_n)$. Hence we define the n**-th homology group**,*

$$H_n := \ker(\partial_n) / \operatorname{im}(\partial_{n+1})$$

*where elements of $\ker(\partial_n)$ are **cycles** and elements of $\operatorname{im}(\partial_{n+1})$ are **boundaries**. The n**-th simplicial homology** of X is $H_n^\Delta(X) = \ker(\partial_n) / \operatorname{im}(\partial_{n+1})$.*

Example 10.1. *Let $X = S^1$. There is one 0-simplex v and one 1-simplex e. Then,*

$$\Delta_0(S^1) = \mathbb{Z}\langle v \rangle,\ \Delta_1(S^1) = \mathbb{Z}\langle e \rangle,\ \Delta_2(S^1) = 0$$

And we have,

$$0 \xrightarrow{\partial_2} \Delta_1(S^1) \xrightarrow{\partial_1} \Delta_0(S^1) \xrightarrow{\partial_0} 0$$

with

$$H_0^\Delta(S^1) = \frac{\ker(\partial_0)}{\operatorname{im}(\partial_1)} = \frac{\mathbb{Z}\langle v \rangle}{\{\sigma_1 \mid_{[v_1]} - \sigma_1 \mid_{[v_0]}\}} = \frac{\mathbb{Z}\langle v \rangle}{\{0\}} \cong \mathbb{Z}\langle v \rangle \cong \mathbb{Z},$$

$$H_1^\Delta(S^1) = \frac{\ker(\partial_1)}{\operatorname{im}(\partial_2)} = \frac{\mathbb{Z}\langle v \rangle}{\{0\}} \cong \mathbb{Z}\langle v \rangle \cong \mathbb{Z},$$

and

$$H_2^\Delta(S^1) = \frac{\ker(\partial_2)}{\operatorname{im}(\partial_3)} = 0$$

We conclude that $H_n^\Delta(S^1) \cong \mathbb{Z}$ if $n = 0, 1$ and $H_n^\Delta(S^1) = 0$ otherwise.

Example 10.2.

$$\begin{tikzcd}
v_0 \arrow[r, "a"] & v_0 \\
v_0 \arrow[u, "b"] \arrow[r, "a"'] \arrow[ur, "c", "L"{near start}, "U"'{near end}] & v_0 \arrow[u, "b"']
\end{tikzcd}$$

We have,

- $\Delta_0(T^2) = \mathbb{Z}\langle v_0 \rangle$
- $\Delta_1(T^2) = \mathbb{Z}\langle a \rangle \oplus \mathbb{Z}\langle b \rangle \oplus \mathbb{Z}\langle c \rangle$
- $\Delta_2(T^2) = \mathbb{Z}\langle U \rangle \oplus \mathbb{Z}\langle L \rangle$
- $\Delta_3(T^2) = 0$

$$0 \xrightarrow{\partial_3} \Delta_2(T^2) \xrightarrow{\partial_2} \Delta_1(T^2) \xrightarrow{\partial_1} \Delta_0(T^2) \xrightarrow{\partial_0} 0$$

The homologies are,

- $H_0^{\Delta}(T^2) = \frac{\ker(\partial_0)}{\operatorname{im}(\partial_1)} = \frac{\mathbb{Z}\langle v_0 \rangle}{\{0\}} \cong \mathbb{Z}\langle v_0 \rangle \cong \mathbb{Z}$
- $H_1^{\Delta}(T^2) = \frac{\ker(\partial_1)}{\operatorname{im}(\partial_2)} = \frac{\mathbb{Z}\langle a \rangle \oplus \mathbb{Z}\langle b \rangle \oplus \mathbb{Z}\langle c \rangle}{\mathbb{Z}\langle a+b-c \rangle} \cong \frac{\langle a,b,c \rangle}{\langle a+b-c \rangle} \cong \langle a, b \rangle \cong \mathbb{Z}\langle a \rangle \oplus \mathbb{Z}\langle b \rangle$
- $H_2^{\Delta}(T_2) = \frac{\ker(\partial_2)}{\operatorname{im}(\partial_3)} = \frac{\mathbb{Z}\langle U-L \rangle}{0} \cong \mathbb{Z}\langle U - L \rangle \cong \mathbb{Z}$
- $H_n^{\Delta} = 0$ *if* $n \geq 3$.

Example 10.3. *Compute* $H_n^{\Delta}(\mathbb{RP}^2)$.

$$\begin{tikzcd}
v_0 & v_1 \arrow[l, "b"'] \arrow[d, "a"] \\
v_1 \arrow[u, "a"] \arrow[r, "b"'] \arrow[ur, "c", "U"{near start}, "L"'{near end}] & v_0
\end{tikzcd}$$

We have,

- $\Delta_0(\mathbb{RP}^2) = \mathbb{Z}\langle v_0 \rangle \oplus \mathbb{Z}\langle v_1 \rangle$
- $\Delta_1(\mathbb{RP}^2) = \mathbb{Z}\langle a \rangle \oplus \mathbb{Z}\langle b \rangle \oplus \mathbb{Z}\langle c \rangle$
- $\Delta_2(\mathbb{RP}^2) = \mathbb{Z}\langle U \rangle \oplus \mathbb{Z}\langle L \rangle$

$$0 \xrightarrow{\partial_3} \Delta_2(\mathbb{RP}^2) \xrightarrow{\partial_2} \Delta_1(\mathbb{RP}^2) \xrightarrow{\partial_1} \Delta_0(\mathbb{RP}^2) \xrightarrow{\partial_0} 0$$

The homologies are,

- $H_0^{\Delta}(\mathbb{RP}^2) = \frac{\ker(\partial_0)}{\operatorname{im}(\partial_1)} = \frac{\mathbb{Z}\langle v_0 \rangle \oplus \mathbb{Z}\langle v_0 \rangle}{\langle v_0 - v_1 \rangle} \cong \mathbb{Z}\langle v_0 \rangle \cong \mathbb{Z}$
- $H_1^{\Delta}(\mathbb{RP}^2) = \frac{\ker(\partial_1)}{\operatorname{im}(\partial_2)} = \frac{\mathbb{Z}\langle a-b \rangle \oplus \mathbb{Z}\langle c \rangle}{\mathbb{Z}\langle a-b-c, a-b+c \rangle}$

  Let $x = a - b$, $y = c$. *Then we have,*

$$x + y = 0, \quad x - y = 0 \implies 2x = 2y = 0$$

i.e. both x and y have order 2. And,

$$x - y = 0 \implies x = y$$

So,

$$H_1^{\Delta}(\mathbb{RP}^2) = \frac{\mathbb{Z}\langle a-b\rangle \oplus \mathbb{Z}\langle c\rangle}{\mathbb{Z}\langle a-b-c,\, a-b+c\rangle} \cong \frac{\mathbb{Z}\langle x\rangle}{2x} \cong \mathbb{Z}/2\mathbb{Z}$$

- $H_2^{\Delta}(\mathbb{RP}^2) = \frac{\ker(\partial_2)}{\mathrm{im}(\partial_3)} = \frac{\{0\}}{\{0\}} \cong \{0\}$
- $H_n^{\Delta} = 0$ *if* $n \geq 3$.

Definition 10.12. *A **singular n-simplex** in a space X is a map $\sigma : \Delta^n \to X$, where there are no conditions of $\mathrm{im}(\sigma)$ restricted to a face (i.e. singularities are allowed). The only requirement is that σ is continuous.*

Definition 10.13. *A **chain complex** $C_n(X)$ is the free abelian group with the set of singular n-simplices in X as basis. Its elements are called n-**chains** which are (finite) formal sum of singular n-simplices.*

$$C_n(X) := \left\{ \sum_i n_i \sigma_i \mid \sigma_i : \Delta^n \to X \right\}$$

with the differential $\partial_n : C_n(X) \to C_{n-1}(X)$ where,

$$\partial_n(\sigma) := \sum_i (-1)^i \sigma \mid_{[v_0,\dots,\widehat{v_i},\dots,v_n]}$$

*and $\partial_n \circ \partial_{n+1} = 0$. The n**-th singular homology** is*

$$H_n(X) := \frac{\ker(\partial_n)}{\mathrm{im}(\partial_{n+1})}$$

Remark 10.14. *$H_n(X)$ is an invariant under the homeomorphism class of X.*

Proposition 10.15. *Let $X = \sqcup X_\alpha$ then $H_n(X) \cong \oplus_\alpha H_n(X_\alpha)$, where each X_α is a distinct path connected component of X.*

Proof. We have $C_n(X) \cong \oplus_\alpha C_n(X_\alpha)$ and $\partial_n = \oplus_\alpha \partial_n^\alpha$ which implies

$$H_n(X) \cong \oplus_\alpha H_n(X_\alpha)$$

□

Proposition 10.16. *If X is non-empty and path-connected then $H_0(X) \cong \mathbb{Z}$.*

Proof. By definition,

$$H_0(X) \cong \frac{C_0(X)}{\mathrm{im}(\partial_1)}$$

We define a homomorphism

$$\varepsilon : C_0(X) \to \mathbb{Z}, \quad \sum_i n_i \sigma_i \mapsto \sum_i n_i$$

This map is surjective since for any $n \in \mathbb{Z}$, $\varepsilon(n\sigma_0) = n$ where $\sigma_0 : \Delta^0 \to X$.

Claim: $\ker(\varepsilon) = \mathrm{im}(\partial_1)$.

Proof. Note that $\text{im}(\partial_1) \subseteq \ker(\varepsilon)$ since for a singular 1-simplex $\sigma : \Delta^1 \to X$ we have $\varepsilon(\partial_1(\sigma)) = 0$.

Now for $\ker(\varepsilon) \subseteq \text{im}(\partial_1)$, let $\varepsilon\left(\sum_i n_i\sigma_i\right) = 0 \implies \sum_i n_i = 0$.

Choose paths $\tau : I \to X$ for $i > 0$, from a base point $x_0 \in X$ to $\sigma_i(v_0)$ and define $\sigma_0(v_0) = x_0$. We can view $\tau_i : [v_0, v_i] \to X$ as a singular 1-simplex. Thus, $\delta(\tau_i) = \sigma_i(v_0) - x_0$. Hence,

$$\partial\left(\sum_i n_i\tau_i\right) = \sum_i n_i\sigma_i(v_0) - \sum_i n_i \cdot x_0 = \sum_i n_i\sigma_i(v_0) \implies \sum_i n_i\sigma_i \in \text{im}(\partial_i)$$

Thus $\ker(\varepsilon) = \text{im}(\partial_1)$. □

That is,

$$H_0(X) \cong \frac{C_0(X)}{\text{im}(\partial_1)} = \frac{C_0(X)}{\ker(\varepsilon)} \cong \mathbb{Z}$$

□

Proposition 10.17. *If X is a point, then $H_n(X) \cong 0$ for $n > 0$ and $H_0(X) \cong \mathbb{Z}$.*

Proof. Let x_0 be a point. Then the constant map $\sigma_n : \Delta^n \to X$ gives us a n-simplex for all n since constant maps are continuous. Since the restriction of the constant map is the constant map we get,

$$\partial_n(\sigma_n) = \sum_{i=0}^{n}(-1)^i\sigma_{n-1} = \sigma_{n-1}\sum_{i=0}^{n}(-1)^i = \begin{cases} 0, & \text{n odd} \\ \sigma_{n-1}, & \text{n even} \end{cases}$$

Thus we get,

$$\cdots \xrightarrow{\partial_3} C_2(x_0) \xrightarrow{\partial_2} C_1(x_0) \xrightarrow{\partial_1} C_0(x_0) \xrightarrow{\partial_0} 0$$

$$\cdots \xrightarrow{0} \mathbb{Z} \xrightarrow{\cong} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0$$

Then for odd n we have, $\ker(\partial_n) = \mathbb{Z}$ and $\text{im}(\partial_{n+1}) = \text{im}(\text{id}_{C_{n-1}}) = \mathbb{Z}$, so

$$H_n(x_0) = \frac{\ker(\partial_n)}{\text{im}(\partial_{n+1})} = \frac{\mathbb{Z}}{\mathbb{Z}} = \{0\}$$

For even $n > 0$, we have $\ker(\partial_n) = 0$, so

$$H_n(x_0) = \frac{\ker(\partial_n)}{\text{im}(\partial_{n+1})} = \{0\}$$

For $n = 0$, we have,

$$H_0(x_0) = \frac{\ker(\partial_0)}{\text{im}(\partial_1)} = \frac{\mathbb{Z}}{0} = \mathbb{Z}$$

Hence, the singular homology groups of a point are 0 for $n > 0$ and $\mathbb{Z}$ for $n = 0$. □

Definition 10.18. *We will change the definition of homology so that a point has trivial $H_0(-)$. The **reduced homology groups** $\widetilde{H_n}(X)$ is defined to be the homology of the following chain complex.*

$$\to C_n(X) \xrightarrow{\partial_n} C_{n-1}(X) \xrightarrow{\partial_{n-1}} \cdots \xrightarrow{\partial_1} C_0(X) \xrightarrow{\varepsilon} \mathbb{Z} \xrightarrow{0} 0$$

So, $\widetilde{H_n}(X) \cong H_n(X)$ if $n \geq 1$. Moreover we have $\operatorname{im}(\partial_1) \subseteq \ker(\varepsilon)$. So,

$$H_0(X) = \frac{C_0(X)}{\operatorname{im}(\partial_1)} \cong \frac{\ker(\varepsilon) \oplus \mathbb{Z}}{\operatorname{im}(\partial_1)} = \widetilde{H_0}(X) \oplus \mathbb{Z}$$

Hence, $\widetilde{H_0}(pt) = 0$.

Definition 10.19. *If $\alpha \in C_n(X)$ we say that α is a **chain**. α is a **cycle** if $\partial_n(\alpha) = 0$, and α is a **boundry** if $\alpha = \partial(\beta)$ for some $\beta \in C_{n+1}(X)$.*

Definition 10.20. *A map $\phi : C_n(X) \to C_n(Y)$ is called a **chain map** if $\partial(\phi) = \phi(\partial)$. Such a ϕ always induces a map in homology.*

Proposition 10.21 (Homotopy Invariance of Singular Homology)**.** *Let $f : X \to Y$ then $\exists$ an induced homomorphism defined by*

$$f_{\#} : C_n(X) \to C_n(Y), \quad \sigma \mapsto f \circ \sigma$$

which extends $\mathbb{Z}$-linearly. This map satisfies $f_{\#}\partial = \partial f_{\#}$.

Proof. For some $\sigma \in C_n(X)$.

$$\begin{aligned} f_{\#}\partial(\sigma) &= f_{\#}\left(\sum_i (-1)^i \sigma|[v_0, \dots, \widehat{v_i}, \dots, v_n]\right) \\ &= \sum_i (-1)^i (f \circ \sigma) \mid_{[v_0, \dots, \widehat{v_i}, \dots, v_n]} \\ &= \partial f_{\#}(\sigma) \end{aligned}$$

Hence $f_{\#}$ induces a map $f_\star : H_n(X) \to H_n(Y)$.

Proof. Note $f_{\#}(\ker \partial_{n_X}) \subset \ker \partial_{n_Y}$. Since $\alpha \in C_n(X)$ such that $\partial(\alpha) = 0$. Then,

$$f_{\#}(\partial_X(\alpha)) = 0 = \partial(f_{\#}(\alpha)) \implies f_{\#}(\alpha) \in \ker(\partial_{n_Y})$$

Again note that $f_{\#}(\operatorname{im} \partial_{n+1_X}) \subset \operatorname{im} \partial_{n+1_Y}$. Let $\alpha \in C_n(X)$ and $\beta \in C_{n+1}(\beta)$ such that $\alpha = \partial(\beta)$. Then,

$$f_{\#}(\alpha) = 0 = f_{\#}\partial(\beta) = \partial(f_{\#}\beta) \implies f_{\#}(\alpha) \in \operatorname{im}(\partial_{n+1})$$

Since $f_{\#}$ sends $\ker(\partial_n)$ to $\ker(\partial_n)$ and $\operatorname{im}(\partial_{n+1})$ to $\operatorname{im}(\partial_{n+1})$ it induces a map $f_\star : H_n(X) \to H_n(Y)$. ☐

☐

Theorem 10.22. *If $f, g : X \to Y$ are homotopic, then they induce the same map on homology,*

$$f_\star = g_\star : H_n(X) \to H_n(Y)$$

Proof. The idea is to subdivide $\Delta^n \times I$ into $(n+1)$-simplicies. Let's say $\Delta^n \times \{0\} = [v_0, \ldots, v_n]$, $\Delta^n \times \{1\} = [w_0, \ldots, w_n]$. We start with $[v_0, \ldots, v_n]$ and consider the $(n+1)$-simplex $[v_0, v_n, w_n]$. Here we keep track of a distinguished edge from $v_n \to w_n$. Note that by contracting along the distinguished edge we go from the $[v_0, \ldots v_n]$-face to $[v_0, \ldots, v_{n-1}, w_n]$-face of the $(n+1)$-simplex. We say that $[v_0, \ldots v_n]$ is the lower face and $[v_0, \ldots, v_{n-1}, w_n]$ the upper face.

In a typical step, we will construct a $(n+1)$-simplex $[v_0, \ldots, v_i, w_i, \ldots, w_n]$ with lower face $[v_0, \ldots, v_i, w_{i+1}, \ldots, w_n]$ and upper face $[v_0, \ldots, v_{i-1}, w_i, \ldots, w_n]$ which we get by contracting along the edge from v_i to w_i. $\square$

Proposition 10.23. *Now given $F : X \times I \to Y$ a homotopy we can define for any singular n-simplex $\sigma : \Delta^n \to X$ the map,*

$$F' : F \circ (\sigma \times \mathrm{Id}) : \Delta^n \times I \to Y$$

*Now define the **prism operator** $P : C_n(X) \to C_{n+1}(Y)$ by*

$$P(\sigma) = \sum_i (-1)^i F' \mid_{[v_0, \ldots, v_i, w_i, \ldots, w_n]}$$

It will turn out that P is not a chain map but

$$\partial P = g_\# - f_\# - P\partial$$

Proof. Let $\sigma : \Delta^n \to X$. Then,

$$\partial P(\alpha) = \sum_{j \le i} (-1)^i (-1)^j F' \mid_{[v_0, \ldots, \widehat{v_j}, \ldots, v_i, w_i, \ldots, w_n]} + \sum_{j \ge i} (-1)^i (-1)^{j+1} F' \mid_{[v_0, \ldots, v_i, w_i, \ldots, \widehat{v_j}, \ldots, w_n]}$$

all terms of $\partial P(\sigma)$ with $i = j$ will cancel except $F' \mid_{[\widehat{v_0}, w_0, \ldots, w_n]} = g \circ \sigma = g_\#(\sigma)$ and $-F' \mid_{[v_0, \ldots, v_n, \widehat{w_n}]} = -f \circ \sigma = -f_\#(\sigma)$ that is $\partial P(\sigma) = g_\# \sigma - f_\# \sigma$ which is a homotopy between $g_\#$ and $f_\#$. The terms with $i \neq j$ are exactly $-P\partial(\sigma)$, that is,

$$P\partial(\sigma) = \sum_{j < i} (-1)^{i-1} (-1)^j F' \mid_{[v_0, \ldots, \widehat{v_j}, \ldots, v_i, w_i, \ldots, w_n]} + \sum_{j > i} (-1)^i (-1)^j F' \mid_{[v_0, \ldots, v_i, w_i, \ldots, \widehat{v_j}, \ldots, w_n]}$$

Thus,

$$\partial P = g_\# - f_\# - P\partial$$

$\square$

Corollary 10.23.1. $\widetilde{H}_i(D^n) = 0$ *for all i.*

11. Exact Sequences

Theorem 11.1. *Let X be a space and A be a subspace of X such that A is closed and it is a deformation retraction of a neighborhood of A in X. Then we have a long exact sequence given by*

$$\ldots \to \widetilde{H_n}(A) \xrightarrow{i_\star} \widetilde{H_n}(X) \xrightarrow{j_\star} \widetilde{H_n}(X/A) \xrightarrow{\partial} \widetilde{H_{n-1}}(A) \xrightarrow{i_\star} \widetilde{H_{n-1}}(X) \xrightarrow{j_\star} \widetilde{H_{n-1}}(X/A) \xrightarrow{\partial} \ldots$$

where $i_\star$ is induced from $i : A \hookrightarrow X$ and $j_\star$ is induced from $j : X \to X/A$.

Definition 11.2. *If (X, A) satisfies the hypotheses of the previous definition of LES, say (X, A) is "good".*

Remark 11.3. *Any CW pair (complex, subcomplex) (X, A) is good, i.e. admits a LES. (Hatcher for proof.)*

Corollary 11.3.1. *$\widetilde{H}_i(S^n) \cong \mathbb{Z}$ if $i = n$, and $= 0$ otherwise.*

Proof. Consider the LES of the pair (D^n, S^{n-1}). Since $D^n/S^{n-1} \cong S^n$, the LES becomes

$$\widetilde{H}_i(S^{n-1}) \to \widetilde{H}_i(D^n) \xrightarrow{0} \widetilde{H}_i(S^n) \to \widetilde{H}_{i-1}(S^{n-1}) \to \widetilde{H}_i(D^n)$$

Since D^n is contractible, $\widetilde{H}_i(D^n) = 0$ for all i, so:

$$0 \to \widetilde{H}_i(S^n) \xrightarrow{\partial} \widetilde{H}_{i-1}(S^{n-1}) \to 0$$

Thus $H_i(S^n) \cong H_{i-1}(S^{n-1})$ for all i, and the computation follows by induction. $\square$

Example 11.1. *$H_0(S^0) \cong \mathbb{Z} \oplus \mathbb{Z}$, since S^0 has 2 path-connected components, so $\widetilde{H}_0(S^0) = \mathbb{Z}$.*

Corollary 11.3.2. *S^n is not homeomorphic to S^m when $n \neq m$.*

Definition 11.4. *Given a space X and a subspace A (where (X, A) is not necessarily good), we define*

$$C_n(X, A) := C_n(X)/C_n(A).$$

Note $\partial : C_n(X) \to C_{n-1}(X)$ sends $C_n(A) \to C_{n-1}(A)$, hence induces a map $\partial : C_n(X, A) \to C_{n-1}(X, A)$.

We get a sequence

$$\cdots \to C_n(X, A) \to C_{n-1}(X, A) \to C_{n-2}(X, A) \to \cdots$$

where $\partial^2 = 0$. We denote the relative homology *by*

$$H_n(X, A) := \ker \partial / \operatorname{im} \partial.$$

Elements of $H_n(X, A)$ are relative cycles*, equivalence classes of $\alpha \in C_n(X)$ such that $\partial\alpha \in C_{n-1}(A)$ (i.e. "$\partial\alpha = 0$"). A relative cycle α is* trivial *if it is a* relative boundary*, i.e. $\alpha = \partial\beta + \gamma$ for some $\beta \in C_{n+1}(X)$, $\gamma \in C_n(A)$.*

TODO:

Theorem 11.5 (Long Exact Sequence of a Pair)**.**

$$\cdots \to H_n(A) \to H_n(X) \to H_n(X, A) \xrightarrow{\partial} H_{n-1}(A) \to \cdots$$

(Proof: homework/reading.)

This follows from the "SES of chain complexes gives rise to LES in homology": given 3 chain complexes (A_n, ∂_n), (B_n, ∂_n), (C_n, ∂_n) (with different notation) and $0 \to C_n(A) \to C_n(B) \to C_n(C) \to 0$ exact for all n, we get

$$\begin{array}{ccccccc} 0 & \to & C_n(A) & \xrightarrow{i_*} & C_n(B) & \xrightarrow{j_*} & C_n(C) \to 0 \\ & & \downarrow \partial & & \downarrow \partial & & \downarrow \partial \\ 0 & \to & C_{n-1}(A) & \xrightarrow{i_*} & C_{n-1}(B) & \xrightarrow{j_*} & C_{n-1}(C) \to 0 \end{array}$$

with $\partial i_* = i_* \partial$ and $\partial j_* = j_* \partial$, and LES:

$$\cdots \to H_n(A) \xrightarrow{i_*} H_n(B) \xrightarrow{j_*} H_n(C) \xrightarrow{\partial} H_{n-1}(A) \xrightarrow{i_*} H_{n-1}(B) \xrightarrow{j_*} H_{n-1}(C) \xrightarrow{\partial} \cdots$$

In our case, the SES is $0 \to C_n(A) \xrightarrow{i_*} C_n(X) \xrightarrow{j_*} C_n(X)/C_n(A) \to 0$, given by inclusion on 8 quotient maps.

Description of ∂: Given $[\alpha] \in H_n(X, A)$, then by definition $\partial\alpha \in C_{n-1}(A)$. Define $\partial[\alpha] = [\partial\alpha] \in H_{n-1}(A)$. Then $\partial(\partial\alpha) = 0$.

Example 11.2. $H_n(X, x_0) \cong \widetilde{H}_n(X)$ *for any* $x_0 \in X$.

Proof. LES with $X = X$, $A = x_0$; since x_0 is reduced, $H_n(x_0) = 0$, so

$$\widetilde{H}_n(X) \cong \widetilde{H}_n(X, x_0).$$

Then $H_n(X, A) \cong \widetilde{H}_n(X, A)$ implies $\widetilde{H}_n(X) \cong H_n(X, x_0)$. □

Theorem 11.6 (Excision). *Let $A, B \subseteq X$ be subspaces whose interiors cover X. Then the inclusion $(B, A \cap B) \hookrightarrow (X, A)$ induces an isomorphism*

$$H_n(B, A \cap B) \cong H_n(X, A) \quad \textit{for all } n \in \mathbb{N}.$$

Idea: One can define a chain complex $C_n(A + B)$ as the subgroup generated by chains in $C_n(X)$ which can be written as a sum of a chain in A and a chain in B. The idea is that $C_n(A + B) \hookrightarrow C_n(X)$ induces an isomorphism in homology. (See Hatcher for proof — "supposedly beautiful.")

Theorem 11.7. *If nonempty open sets $U \subseteq \mathbb{R}^n$, $V \subseteq \mathbb{R}^m$ are homeomorphic, then $n = m$.*

Proof. For $x \in U$, consider $\mathbb{R}^n = U \cup \mathbb{R}^n \setminus \{x\}$. Then by excision,

$$H_k(U, U \setminus \{x\}) \cong H_k(\mathbb{R}^n, \mathbb{R}^n \setminus \{x\}) \quad \forall k.$$

Compute $H_k(\mathbb{R}^n, \mathbb{R}^n \setminus \{x\})$ via the LES:

$$\to H_k(\mathbb{R}^n \setminus \{x\}) \to H_k(\mathbb{R}^n) \to H_k(\mathbb{R}^n, \mathbb{R}^n \setminus \{x\}) \to H_{k-1}(\mathbb{R}^n \setminus \{x\}) \to H_k(\mathbb{R}^n)$$

Since $\mathbb{R}^n \setminus \{x\} \simeq S^{n-1}$ and $\mathbb{R}^n$ is contractible:

$$H_k(S^{n-1}) \to 0 \to H_k(\mathbb{R}^n, \mathbb{R}^n \setminus \{x\}) \xrightarrow{\cong} H_k(S^{n-1}) \to 0$$

So $H_k(\mathbb{R}^n, \mathbb{R}^n \setminus \{x\}) \cong H_k(S^{n-1}) \cong \mathbb{Z}$ if $k = n$, and 0 otherwise.

If $f : U \to V$ is a homeomorphism, then f induces $f_* : (U, U \setminus \{x\}) \xrightarrow{\text{homeo}} (V, V \setminus \{f(x)\})$, so

$$f_* : H_k(U, U \setminus \{x\}) \to H_k(V, V \setminus \{f(x)\})$$

is an isomorphism, giving $H_k(\mathbb{R}^n, \mathbb{R}^n \setminus \{x\}) \cong H_k(\mathbb{R}^m, \mathbb{R}^m \setminus \{f(x)\})$.

The LHS is $\mathbb{Z}$ when $k = n$ and 0 otherwise; the RHS is $\mathbb{Z}$ when $k = m$ and 0 otherwise. Therefore $n = m$. $\square$

12. Equivalence of Simplicial and Singular Homology

Put a Δ-complex structure on X, then show $H_n^\Delta(X) \cong H_n(X)$.

To show this, we need the following definition.

Definition 12.1. *If $A \subseteq X$ is a subspace, the* ***relative simplicial chain complex*** *is*

$$\Delta_n(X, A) := \Delta_n(X)/\Delta_n(A),$$

with $\partial : \Delta_n(X) \to \Delta_{n-1}(X)$ inducing $\partial : \Delta_n(X, A) \to \Delta_{n-1}(X, A)$. Define

$$H_n^\Delta(X, A) := \ker \partial / \operatorname{im} \partial.$$

We get a SES $0 \to \Delta_n(A) \to \Delta_n(X) \to \Delta_n(X, A) \to 0$ and hence a LES in homology $H_n^\Delta \quad (*)$.

Recall $\Delta_n(X) = \{\text{maps } \Delta^n \to X\}$ with some nice properties, and $C_n(X) = \{\text{maps } \Delta^n \to X\}$. There is a canonical map $\phi : \Delta_n(X) \to C_n(X)$, $(\Delta^n \to X) \mapsto (\Delta^n \to X)$, forgetting the nice properties.

Claim: ϕ induces an isomorphism $H_n^\Delta(X) \xrightarrow{\sim} H_n(X)$.

Today we prove this claim for the case where X is finite-dimensional, i.e. $X^k = \varnothing$ for all $k \geq k_0$ (X^k is the k-skeleton, using simplices of dimension $\leq k$).

Proof. We have a commutative diagram with LES rows:

$$\begin{array}{ccccccccccc}
\to H_{n+1}^\Delta(X^k, X^{k-1}) & \to & H_n^\Delta(X^{k-1}) & \to & H_n^\Delta(X^k) & \to & H_n^\Delta(X^k, X^{k-1}) & \to & H_{n-1}^\Delta(X^{k-1}) \to \\
\downarrow \phi_* & & \downarrow \phi_* & & \downarrow \phi_* & & \downarrow \phi_* & & \downarrow \phi_* \\
\to H_{n+1}(X^k, X^{k-1}) & \to & H_n(X^{k-1}) & \to & H_n(X^k) & \to & H_n(X^k, X^{k-1}) & \to & H_{n-1}(X^{k-1}) \to
\end{array}$$

We show maps ① and ④ are isomorphisms.

Note $\Delta_n(X^k, X^{k-1}) \cong \Delta_n(X^k)/\Delta_n(X^{k-1}) = \begin{cases} \Delta_n(X^k) & \text{if } n = k \\ 0 & \text{otherwise} \end{cases}$.

Now if $d \in \Delta_n(X^n)$, then $\partial d \in \Delta_{n-1}(X^{n-1})$ by face properties, so $\partial_{\text{rel}} = \Delta_n(X^n) \to 0$ implies $\operatorname{Im} \partial_{\text{rel}} = 0$. Since $\partial(\cdot) = 0$, $H_n^\Delta(X^k, X^{k-1})$ is free abelian for $n = k$ and 0 otherwise.

Now consider $\bigsqcup_\alpha \Delta_\alpha^k / \bigsqcup_\alpha \partial \Delta_\alpha^k \xrightarrow{\sim} (X^k, X^{k-1})$.

But $\bigsqcup_\alpha \Delta_\alpha^k / \bigsqcup_\alpha \partial \Delta_\alpha^k \cong X^k / X^{k-1}$ by quotient map / homeomorphism ($\cong$ means homeomorphism here).

n**-simplices** (quotient by boundary): $\bigvee_\alpha S^k := \{\bigsqcup_\alpha S^k / x_\alpha \in S_\alpha^k \times \{\beta\}, \forall \alpha, \beta\}$ (*wedge product*), so:

$$H_n\left(\bigvee_\alpha S^k\right) \cong \bigoplus_\alpha H_n(S^k) \to \text{free abelian for } n = k,\ 0 \text{ otherwise.}$$

So the domains/codomains of ① and ④ in the diagram are isomorphic. Why, induced by ϕ, though? Because ϕ is almost the "identity" map; the isomorphism is the "same" map, so we use it.

By the Five Lemma (applied to maps ② and ③ isomorphisms), we get ② and ⑤ isomorphisms. The five lemma says: if you have an exact sequence

$$A \to B \to C \to D \to E, \qquad A \to A',\ B \to B',\ D \to D',\ E \to E' \text{isomorphisms},$$

then $C \to C'$ is also an isomorphism.

Repeat until $X = X^{k_0}$, and remaining things are 0. Done. □

Used in proof:

Theorem 12.2. *If (X, A) is a good pair, then $q : (X, A) \to (X/A, A/A)$ induces an isomorphism*

$$q_* : H_n(X, A) \to H_n(X/A, A/A) \cong H_n(X/A, \{p\}) \cong \widetilde{H}_n(X/A).$$

This gives $H_n^\Delta(X^k, X^{k-1}) \cong H_n^\Delta(X^k/X^{k-1})$. Then since $X^k/X^{k-1} \cong \bigvee S^k$, use the LES of $(\bigsqcup S^k, \bigsqcup\{x_\alpha\})$ and $\widetilde{H}_n(\{p\}) = 0$:

$$\widetilde{H}_n\left(\bigsqcup S^k, \bigsqcup x_\alpha\right) \cong \widetilde{H}_n\left(\bigsqcup S^k\right) \cong \bigoplus_\alpha H_n(S^k).$$

Mayer–Vietoris Sequence. Suppose $X = A \cup B$, with $C_n(A + B) \hookrightarrow C_n(X)$.

Claim: there is a SES

$$0 \to C_n(A \cap B) \xrightarrow{\phi} C_n(A) \oplus C_n(B) \xrightarrow{\psi} C_n(A + B) \to 0$$

where $\phi(x) = (x, -x)$ and $\psi(x, y) = x + y$.

If we prove it's exact, we'll get a LES in homology.

- ϕ is injective since $\phi(x) = 0 \Rightarrow x = 0$.
- $\ker \psi = \{(x, y) \mid x + y = 0\} = \{(x, -x) \mid x \in C_n(A), -x \in C_n(B)\} = \{(x, -x) \mid x \in C_n(A \cap B)\} = \operatorname{im} \phi$.
- ψ is surjective (from definition of $C_n(A + B)$).

Thus the sequence is exact, so we get a LES in homology, called the **Mayer–Vietoris sequence**:

$$\cdots \to H_n(A \cap B) \to H_n(A) \oplus H_n(B) \xrightarrow{\psi_*} H_n(A + B) \xrightarrow{\partial} H_{n-1}(A \cap B) \to \cdots$$

(also has a reduced version.)

Note: If $A \cap B$ is path-connected, then

$$\widetilde{H}_1(A \cap B) \xrightarrow{\phi_*} \widetilde{H}_1(A) \oplus \widetilde{H}_1(B) \xrightarrow{\psi_*} \widetilde{H}_1(X) \to \widetilde{H}_0(A \cap B)^0$$

then $H_1(X) \cong H_1(A) \oplus H_1(B)/\ker \psi_* \cong H_1(A) \oplus H_1(B)/\operatorname{im} \phi_*$. (Similar to Van Kampen's theorem.)

Remark 12.3. *H_1 is the abelianization of π_1. In particular, $H_1(\Sigma_g) \cong \pi_1(\Sigma_g)^{\text{ab}} \cong \mathbb{Z}^{2g}$.*

Suspension of a Space. Given a space X, its *suspension* is

$$\Sigma X := X \times [-1,1] \ / \ (x_1 \sim x_2,\ X \times \{1\},\ X \times \{-1\} \text{ for } x_1, x_2 \in X).$$

Take A = upper cone $\cong$ cone on X, B = lower cone = "slightly extended cone"; then $A \cup B = \sum X$ and $\widetilde{H}_n(A) \cong \widetilde{H}_n(B) \cong 0$ since any cone on X is contractible. So by Mayer–Vietoris:

$$\widetilde{H}_n(\Sigma X) \cong \widetilde{H}_{n-1}(A \cap B) \cong \widetilde{H}_{n-1}(X) \quad \text{by definition / reduction.}$$

Definition 12.4. ***Degree*** *of a map $f : S^n \to S^n$ is $f_\star(1) \in \mathbb{Z}$ where $f_\star : H_n(S^n) \to H_n(S^n)$ and $1 \in \mathbb{Z} \cong H_n(S^n)$.*

Example 12.1.

(1) $(\mathrm{Id})_\star = \mathrm{id} \implies \deg \mathrm{Id} = 1$
(2) *If $f : S^n \to S^n$ is not a surjection then* $\deg f = 0$ *(potential final problem).* ***TODO:*** *For f we can assume that f misses a pt $\in S^n$ so we can factor f as a map $S^n \xrightarrow{\overline{f}} S^n \setminus \{x_0\} \hookrightarrow S^n$ so $f_\star = (i \circ \overline{f})_\star = (i_\star)(\overline{f})_\star$ so* $\deg = 0$
(3) *If $f \simeq g$ then* $\deg f = \deg g$.
(4) $\deg(f \circ g) = \deg f \times \deg g$
(5) *If f is a homotopy equivalence then* $\deg f = \pm 1$.
(6) *If f is a reflection along a equitorial sphere then* $\deg f = -1$. *(Hatcher, Chapter 2.1)*
(7) *If f is the anti-podal map on S^n then* $\deg f = (-1)^{n+1}$.
(8) *If $f : S^n \to S^n$ has no fixed points then* $\deg f = (-1)^{n+1}$.

Proof. If $f(x) \neq x$ for all $x \in S^n$ then the line segment joining $-x$ to $f(x)$ does not pass through the origin. Then,

$$f_t(x) = \frac{(1-t)f(x) + t(-x)}{|(1-t)f(x) + t(-x)|}$$

defines a homotopy of f to the antipodal map in S^n. So, $f_t : S^n \to S^n$. $\square$

Theorem 12.5. *S^n has a non-zero vector field if and only if n is odd.*

Proof. ($\implies$)

Vector field in S^n corresponds to $S^n \xrightarrow{v} \mathbb{R}^{n+1}$, $x \mapsto v(x)$ such that $x \perp v(x) \iff \langle x, v(x) \rangle = 0$. Suppose v is such a vector field on S^n. We may normalize $v(X)$ by $\frac{v(x)}{|v(x)|}$ so that v can be thought of as a map $S^n \xrightarrow{v} S^n$. Now consider $\cos(t) \cdot x + \sin(t) \cdot v(x)$ By letting t go from 0 to π, we obtain a homotopy $f_t : S^n \to S^n$ from the identity to the antipodal map.

$\implies \deg \mathrm{id} = \deg(-\mathrm{id}) \implies 1 = (-1)^{n+1} \implies n$ is odd.

($\impliedby$)

Suppose n is odd. Say $n = 2k-1$. Define $v(x_1, \ldots, x_{2k}) = (-x_2, x_1, \ldots, -x_{2k}, x_{2k-1}) \in \mathbb{R}^{n+1}$.

$x \perp v(x)$ and $\|v(x)\| = 1$. $\square$

Definition 12.6. ***Euler characteristic*** *of a space* X

$$\chi(X) := \sum_n (-1)^n rank(H_n(X))$$

rank of a finitely-generated, free, abelian group is the number of $\mathbb{Z}$*-summands in that group.*

Example 12.2. *rank*$(Z \otimes Z \otimes Z) = 1$. $\chi(T^2) = (-1)^0(1) + (-1)^1(2) + (-1)^2(1)$